

Succession from Strength: Coups and Autocratic Succession Rules in Sub-Saharan Africa

Sam Sharman*

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Abstract

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*PhD Candidate, Department of Political Science, Texas A&M University; Incoming Assistant Professor, Graduate School of International Relations, International University of Japan. Email: ssharman13@tamu.edu. Address: 3050 Allen Building, 4220 George Bush Dr W, College Station, TX 77843. ORCID: 0000-0002-3065-1301. I thank David Bearce, José Antonio Cheibub, Scott J. Cook, Matthew Fuhrmann, Quan Li, Matt Malis, and Eddy S. F. Yeung, participants at the Midwestern Political Science Association, and seminar audiences at the International University of Japan, Louisiana State University, and Texas A&M University for their comments on previous drafts.

Abstract

Why does a minority of autocracies fail to formalize succession rules in the constitution? According to recent research, weak dictators have succession rules to reduce the threat of coups. In contrast, I argue that dictators have succession rules when they are already secure. Succession rules reduce the threat of succession crises but create potential threats by empowering successors and angering elites. Given the potential threats created by succession rules, dictators will create them when they have sufficiently consolidated power. I use Random Forest, a machine learning algorithm, to predict the probability of successful coups. Using original data on succession rules for 47 sub-Saharan African countries from 1966–2016, I show that dictators are less likely to have succession rules when coups are more likely. Contrary to prevailing approaches to autocratic institutions, succession rules are tools of stable and secure dictators, not threatened ones. The argument and findings have implications for the study of autocratic succession, the role of domestic institutions in international politics, and institutional development more broadly.

Why does a minority of autocracies fail to formally prepare for political succession? Recent research has begun to examine the role of succession rules in non-routine transfers of power, particularly designated successors. A designated successor is an individual or, more commonly, the holder of a specific office named in the constitution to assume power if a leadership vacancy occurs. Having a designated successor presents a simple puzzle. Designated successors can threaten the dictator. The designated successor has the motivation to overthrow the dictator, maximizing the successor's time in power and preventing their early removal from their role as successor. Through their status, the successor gains resources and allies who can facilitate a coup d'état against the dictator.¹ A designated successor has both the means and motive to overthrow the dictator. Herz (1952) named this potential danger from a successor the crown-prince problem.

Despite the crown-prince problem, existing research has reached a consensus that designated successors offer more benefits than costs. Designated successors directly protect dictators from coups. Coup plotters must overthrow both the dictator and the successor, increasing the difficulty of staging coups (Frantz and Stein 2017; Konrad and Mui 2017; Meng 2021). Designated successors also reduce the incentives for other elites to plot coups. Elites may stage coups to ensure a transfer of power if the dictator is vulnerable. But designated successors reduce uncertainty over the future and assure elites that the regime will persist past the current dictator (Kokkonen, Møller, and Sundell 2022; Kokkonen and Sundell 2014; Kurrild-Klitgaard

1. Following Svoblik (2012), I use the term coup to refer to “a forced removal of an authoritarian leader by *any* regime insider, not necessarily the military” (4n8; italics in original).

2000; Tullock 1987). Because designated successors reduce the risk of coups, dictators are likely to establish succession rules when they fear such threats.

This line of research treats succession rules themselves as the puzzle. In reality, succession rules are commonplace: over 77% of autocracies in sub-Saharan Africa have succession rules in my data. The puzzle, therefore, should be reframed in terms of why a minority do *not*. I address this puzzle by reversing the relationship between leader survival and succession rules. Previous work argues that only weak dictators create succession rules by simply assuming that strong dictators do not. For example, in an important contribution, Meng (2021) asserts that “a leader who is *already* secure and anticipates a smooth succession has no reason to create succession policies” (957; italics in original). The relationship between the threat of a coup and the presence of succession rules has not received full consideration, either theoretically or empirically.

The crown-prince problem leaves open an alternative explanation for why dictators have succession rules despite the risk of coups: the dictator does not fear the threats from succession rules, whether from a successor or other elites. If the dictator believes that they can prevent coups, the crown-prince problem is not a concern. Instead, the possibility that strong dictators have succession rules is assumed away. I give this possibility its first serious consideration. With an already secure dictator, succession rules reduce the threat of perilous transfers of power after a vacancy while not threatening the dictator in their lifetime. The uncommon absence of succession rules is explained by autocracies where a dictator never becomes sufficiently secure.

I test the argument using a new strategy for estimating coup probabilities and original data on succession rules in sub-Saharan African autocracies from 1966 to 2016. For this analysis, I focus on sub-Saharan Africa to ensure comparability with previous results and exclude substitutes for succession rules like dominant ruling parties and parliamentary constitutions. The probability of a successful coup is estimated with Random Forest, a machine learning algorithm. The Random Forest probabilities provide an accurate and direct measurement of a dictator’s vulnerability to a coup that is dynamic within a single dictator’s tenure. I then model the relationship between the predicted probability of a coup and succession rules using probit models. When the probability of a coup is higher, autocracies are less likely to have succes-

sion rules. The appendices show that the results are robust to accounting for reverse causality, unobserved heterogeneity, and using coup attempts, and provide an illustrative case study of Zimbabwe.

This paper contributes to the study of autocratic succession, the role of domestic institutions in international politics, and institutional development more broadly. The most obvious implications concern autocratic succession. The prevailing consensus holds that dictators formalize succession rules to prevent coups. By implication, dictators have succession rules when they are weak and vulnerable to coups. Existing work has shown a consistent correlation between the presence of succession rules and the avoidance of coups (Frantz and Stein 2017; Kokkonen and Sundell 2014; Kokkonen, Møller, and Sundell 2022; Konrad and Mui 2017; Kurrild-Klitgaard 2000; Meng 2021; Tullock 1987; Zhou 2023). However, this paper provides evidence that the relationship is, in fact, reversed: dictators have succession rules when they are already strong and can prevent coups. Previous arguments linking succession rules to coup-proofing require retesting. Even if the observed relationship persists, the existing theoretical framework remains incomplete.

The findings, further, speak to a growing body of research on how autocratic institutions shape international outcomes. Scholars have argued that autocratic institutions affect international outcomes like conflict behavior (Weeks 2012), cooperation (Mattes and Rodríguez 2014), and foreign investment (Wright 2008) by constraining dictators and providing information to external actors. Such arguments typically assume that institutional variation reflects variation in leader constraints.

If institutions instead reflect the security of the dictator, as my argument and findings suggest, the information content of institutions for international audiences is different from what existing theories assume. A dictator who adopts succession rules from a secure position is not signaling that they are constrained and may, in fact, be signaling the opposite. This distinction matters for theories that treat autocratic institutions as constraints because the same institution may carry drastically different implications for cooperation, conflict, and investment depending on why the dictator adopted the institution. Understanding the domestic logic behind autocratic institutions is, therefore, prior to evaluating their international consequences.

More broadly, my argument challenges the power-sharing framework that links weakness to institutionalization. Since Geddes (1999), the study of autocracy has emphasized the role of institutions (Pepinsky 2014). Nominally-democratic institutions—such as cabinets, constitutions, elections, legislatures, and parties—are associated with longer-serving dictators and more stable regimes. Autocratic institutions facilitate cooperation between dictators and elites, helping to establish boundaries on dictatorial power and allocate resources and policy control (Arriola 2009; Boix and Svolik 2013; Brownlee 2007; Gandhi 2008; Gandhi and Przeworski 2007; Geddes, Wright, and Frantz 2018; Magaloni 2008; Meng 2020; Svolik 2012). The same logic appears in institutional economics, where representative institutions emerge from state weakness and a need to bargain with powerful social groups. Representative institutions served as venues for these groups to exercise their powers and restrain the state (Levi 1988). These literatures can be unified under a power-sharing framework that links weakness to institutionalization. A weak ruler or sovereign, needing the cooperation of a relatively stronger group to maintain power, sacrifices authority and resources to ensure survival. Institutions are then created to formalize and solidify these agreements (Meng 2020; Meng and Paine 2022; Meng, Paine, and Powell 2023).

The argument and findings of this paper challenge the power-sharing framework in two key respects. First, institutions are central to ordinary politics. They perform essential governance functions, such as organizing succession. As such, institutions do not always need to be explained by their role in prolonging a dictator's tenure. Second, already strong and secure dictators have the greatest capacity to create institutions. According to the power-sharing framework, weak dictators must sacrifice power and resources. Yet weak dictators are weak precisely because they have less power relative to elites and lack sufficient resources. In contrast, strong dictators possess the capacity to redistribute power and resources, enabling them to establish institutions. These points have antecedents in previous work showing more nuanced relationships between power and institutions, such as strong monarchs introducing representative institutions (Boucoyannis 2021), institutionalization bolstering a dictator's ability to implement policies (Slater 2003), and power-sharing being a strategy for dictators to avoid blame (Williamson 2024).

The next section discusses the literature on coups and succession rules in greater depth. Then, I argue that autocracies should be more likely to have succession rules when coups are already unlikely. The remainder of the paper tests whether the probability of a coup affects having succession rules. I start by introducing an original dataset on succession rules in sub-Saharan Africa and using a Random Forest model to predict the probability of a successful coup. Next, I describe the empirical model and discuss the results, finding a negative relationship between the probability of a coup and having succession rules. The final section concludes.

Choosing to Have Succession Rules

Succession rules serve an essential purpose across all types of political regimes. Organizing transfers of power is a basic task of governance. For political regimes to survive in the long term, they must be prepared to transfer power between leaders at any moment. Leader vacancies are always a possibility. Politics does not supersede biology: leaders fall ill, become incapacitated, and die. The fundamental purpose of succession rules is to help regimes navigate and survive leader vacancies.

Although all regimes must handle succession if they endure long enough, succession is a greater threat to autocracies than democracies. Democracies have inherent strategies for succession through elections. If a vacancy occurs, the next leader is ultimately chosen by an election, and the succession rule determines who will take over until the next election and when the next election occurs. Conversely, autocracies lack inherent systems for succession and often fail to prepare for it altogether. Autocracies frequently arise from crises and overturn the existing constitutional order. Elites in new autocratic regimes can elect to ignore establishing formal procedures in the constitution—if they choose to create a constitution at all.

A vacancy in the dictator's position launches a potential power struggle. The dictator personifies the regime, determining policy and the distribution of privileged positions among elites. When a vacancy occurs, ambitious elites will seek to take power themselves. Other elites will begin organizing around their favored candidates. The stakes of autocratic succession are high. At best, losing candidates and their supporters receive fewer benefits. They risk severe pun-

ishments if they failed to support the new dictator, ranging from exclusion from the regime to imprisonment or execution. If elites cannot agree on the next leader or are uncertain about the likely victor in a succession contest, they may stage coups to secure a favorable outcome.

The stakes of succession contests and potentially high levels of uncertainty that they involve ultimately stem from coordination problems. As a form of constitutional rule, succession rules help to resolve coordination problems. The core purpose of constitutions is to facilitate coordination. Constitutions provide rules that coordinate behavior by strengthening mutual expectations and increasing the costs of deviation (Carey 2000; Hardin 1989). While the effects of constitutions are mostly associated with democracies, constitutions serve similarly important roles in autocracies. Modern governments, regardless of regime type, are complex and require significant coordination between leaders and regime members. Constitutions provide the rules that regulate decision-making and interactions between members of government, particularly dictators and elites (Albertus and Menaldo 2012; Barros 2002; Myerson 2008). More broadly, constitutions serve as operating manuals that inform the dictator and elites how the government should function (Przeworski 2013).

Specifically, succession rules reduce the threat of vacancies turning into succession crises and coups by reducing uncertainty over the power struggle's outcomes and establishing structures for bargaining between factions. Without succession planning, a leadership vacancy leaves elites uncertain over the regime's future and the next dictator. The next dictator could emerge as a threat to the interests of a subset of elites. Rather than waiting for the contest to play out, these elites may exploit the power vacuum and stage a coup. A successful coup guarantees an outcome favorable to the plotters rather than risking the emergence of a dictator opposed to their interests. This possibility can be exacerbated by the presence of factions that disagree over candidates for succession. If elites do not believe that they can agree on the next dictator, factions may race to commit coups and prevent rivals from taking power.

Succession rules reduce the risk that uncertainty will motivate coups. Succession rules influence the probability distribution over who will take power in the event of a vacancy. The most obviously advantaged candidate is the successor, if one exists, even if they only assume power on an interim basis. In the event of a vacancy, the successor gains a structural advantage

as holder of the dictator's office and becomes one of the most obvious candidates to coordinate around. Even figures who have ranked as secondary figures in the regime, such as Paul Biya of Cameroon and Daniel arap Moi of Kenya, have consolidated power over the regime by virtue of being the successor during a vacancy.

Most succession rules either do not name successors or only empower successors on an interim basis. Instead, a separate process, such as selection by the legislature or the ruling party, determines the next leader. Although such rules provide less certainty than those with designated successors, they still shape elites' understandings of the future. Elites can form expectations regarding the succession outcome based on the composition of the body responsible for choosing the next leader and the preferences of its members. With succession rules in place, elites develop expectations about the regime's future if the dictator dies, including who the most likely successors will be.

Processes for choosing the next dictator further reduce the threat of factions turning to violence. Violence can emerge among factions when elites are uncertain about reaching a compromise candidate. Succession rules provide a structure and venue for factions to bargain and reach a compromise candidate, increasing confidence that such a candidate can be found and reducing the likelihood of factions resorting to violence. Staging a coup and establishing a new regime are costly and risky. A compromise candidate who preserves the regime is likely preferable to the risks of attempting to establish a new regime through a coup. Processes in succession rules reduce the risk of factionalization leading to coups by providing a venue for selecting a compromise candidate.

In short, succession rules help autocracies avoid succession crises and ensure a peaceful transfer of power in the event of a dictator's death. Succession rules reduce uncertainty by creating shared expectations over the likely outcomes of power struggles. They also mitigate the threat of factional conflict by providing processes that can select a compromise candidate. Given that succession rules are core elements of political regimes and modern constitutions, the more interesting question is why a minority of autocracies lack them. The answer lies in the new threats that succession rules can create.

The costs of succession rules arise from the creation of potential rivals. Succession rules

produce two broad groups within the regime. The advantaged benefit from the succession rule; if the dictator dies and the rule is followed, they are happy with the anticipated outcome. The most obvious member of the advantaged is any named successor, and others include the successor's allies and members of the regime who prize stability. Conversely, the disadvantaged strongly prefer an outcome that diverges from the anticipated effect of the succession rule. Those most disadvantaged are ambitious elites who are not named the successor, their allies, and other enemies of the successor.

The "barrier effect" from existing work can be reinterpreted in terms of the incentives and relative power of these two groups. The advantaged and disadvantaged have opposing incentives. The advantaged want to preserve the regime and wait for the dictator's death so that the succession rule takes effect. The disadvantaged want to stage a coup and install their preferred candidate in power. The balance of power between the advantaged and disadvantaged can help explain when dictators choose to implement succession rules. If the protective power of the advantaged outweighs the threat of the disadvantaged, the succession rule enhances the dictator's overall security. In such cases, a threatened dictator then has an incentive to use succession rules as a survival strategy.

The incentives for the advantaged, however, can be more nuanced. Certainly, the disadvantaged have stronger reasons to consider staging a coup. Appointing a successor cuts off other elites from taking power peacefully. Once the successor is established, a coup may become the only means for a non-successor to assume power. Elites who do not seek the crown themselves still have preferences regarding the next ruler. An ally assuming power could increase an elite's privileges within the regime, while an adversary taking control could lead to imprisonment, torture, or execution. If elites are dissatisfied with the succession rule's expected outcome, a coup provides them an opportunity to eliminate that possibility.

The advantaged, especially the successor, still have incentives to stage a coup rather than simply wait for the dictator's death. Most successors will want to maximize their time in power as the dictator. Successors may stage a coup to depose the dictator early and extend their tenure. Not all successors will have the ability to stage a successful coup when first appointed. However, over time, the successor can consolidate power and increase the likelihood

of a successful coup.

Moreover, the longer the successor waits for the incumbent to leave, the more likely they are to lose their position. Any successor becomes more likely to die as they wait. The successor's growing power also suggests a strategic response from the incumbent. The incumbent may remove the successor before they become too powerful. Regularly rotating successors minimizes the risk of a coup by preventing any successor from amassing sufficient power. Elites are less likely to build alliances with a successor they do not expect to remain in office for long. Staging a coup can help the successor avoid losing their position.

The remaining question is why already secure dictators have succession rules. Once power is consolidated, having a succession rule may be uncontroversial. Succession rules are standard elements of modern constitutions, and having one resolves a basic governance problem. This explanation comports with the core fact that succession rules are the default, suggesting that their absence is the true puzzle. Still, two specific reasons can account for a dictator's motivations for adopting succession rules without assuming concern for the regime's future after their death.

First, succession rules provide information to actors outside the regime. Their presence signals that the dictator is confident that coups can be stopped. Given how commonplace succession rules are, their absence likely sends an even stronger signal that the dictator is vulnerable. While many external groups maintain an interest in a dictator's stability, foreign investors are an obvious group because political instability threatens their investments. Succession rules could signal to investors that coups are unlikely and reassure them that the regime would survive a succession crisis.

Second, succession rules can help dictators manage domestic coalitions. Elites value succession rules because they help the regime survive in the long term (Frantz and Stein 2017). Succession rules also provide tools for patronage and for rewarding coalition members. Autocracies, especially in Africa, rely on cabinets to distribute rents (Arriola 2009; Francois, Rainer, and Trebbi 2015). Succession and cabinets are closely linked. The most common type of successor in my data holds a cabinet post, accounting for 43% of successors. Specifically, 35.8% are vice presidents, 5.56% are prime ministers, and 1.85% are other cabinet ministers.

Introducing a cabinet post for the successor provides the dictator, at a basic level, with an additional post through which to distribute patronage. However, the successor's post comes with additional rewards beyond those of a standard cabinet appointment. Successors have a heightened status and are among the most obvious candidates for succession. Even if the successor could not win an open contest, they possess a substantial structural advantage if they hold the successor position when a vacancy occurs.

However, this function effectively operates as a luxury good. The dictator can use it to reward and appease most elites, but at the cost of angering other elites. A weak dictator cannot use it because they cannot prevent a coup. A strong dictator can use succession rules as a luxury good, substituting material costs, such as providing rents and policy concessions, for a marginal increase in coup risk. Succession rules do not reduce a dictator's baseline vulnerability to coups; instead, they redistribute the costs in ways that only secure dictators can afford.

When dictators have succession rules, they are not merely balancing a group of would-be protectors against a group of would-be coup plotters. Both groups have incentives to attempt coups and seize power. Coups are more likely to succeed against weaker dictators, making successors and elites more inclined to favor coups in such cases. Stronger dictators can deter coups and keep both the advantaged—especially the successor—and the disadvantaged placated. Consequently, dictators are most likely to introduce and retain succession rules after consolidating their rule and ensuring they can prevent coups.

Data on Succession Rules

To test the relationship between the risk of losing power and having succession rules, I collect data on succession rules for all autocracies in sub-Saharan Africa. Sub-Saharan Africa is defined as member states of the African Union that are *not* categorized as Northern Africa by the African Union. Autocracies are identified using an expansive sample of any country coded as autocratic by at least one of Boix, Miller, and Rosato (2012), Cheibub, Gandhi, and Vreeland (2010), and Geddes, Wright, and Frantz (2014). Constitutions are accessed primarily through the Comparative Constitutions Project (Elkins, Ginsburg, and Melton 2014) and World Con-

stitutions Illustrated. In the absence of a permanent constitution, I use the primary governing document where possible, including suspension orders, ordinances, decrees, and fundamental laws.

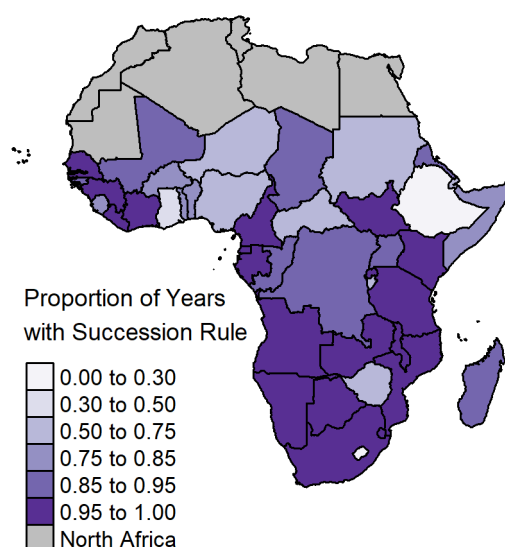
The data address limitations of existing data for succession rules. Frantz and Stein (2017) use Polity's executive regulation variable. Their operationalization includes non-competitive elections as a succession rule, so routine and non-routine procedures are lumped together. Using the Polity variable cannot distinguish whether succession rules for permanent vacancies exist independently of elections. Meng (2021) collects data on succession rules with named successors, but the data only include succession rules in place directly before leadership transitions or at the end of a dictator's tenure. While Meng's data are static, succession rules are dynamic and can change during a single dictator's tenure. Fully characterizing succession rules requires time-series-cross-sectional data rather than a pure cross-section.

Keeping the analysis in sub-Saharan Africa ensures comparability with previous results and controls for important confounders. Existing research on succession rules in modern autocracies has focused on sub-Saharan Africa. The hypothesis proposed in this paper—that succession rules are more likely when the probability of a coup is low—predicts the opposite conclusion from existing work. If the hypothesis is supported, using data from the same region increases confidence that the results contradicting existing work stem from theoretical reasons and not the choice of cases.

Additionally, using sub-Saharan Africa controls for other means of managing succession without formal rules. Both ruling parties and parliamentary governments can substitute for rules. Ruling parties in African autocracies are typically incorporated into the state formally rather than supplanting the constitution. Constitutions establish the head of the party as the head of state, and party organs are responsible for managing the succession process. The power of the ruling party is formalized in sub-Saharan African autocracies. Parliamentary governments are also uncommon. Although many former British colonies adopted parliamentary constitutions immediately after independence, most eventually switched to presidential systems.² While

2. Even the most prominent remaining parliamentary governments in the region, Botswana and South Africa, fuse the head of government and head of state together in an executive presidency rather than separate the roles into a prime minister and president. The president in both systems has a formal line of succession.

Figure 1. Proportion of Years with Succession Rules among Sub-Saharan African Autocracies through 2020



Notes: Countries with darker shades have succession rules in a higher proportion of years that they are autocratic since independence. Countries in grey are categorized as Northern Africa by the African Union and are excluded from the data.

strong ruling parties and parliamentary governments are alternatives to succession rules, they are less of an inferential threat in sub-Saharan Africa.

To construct the succession rules variable, I code whether an autocracy has an identifiable succession rule at a monthly level. Succession rules define how to handle permanent vacancies in the chief executive. A succession rule exists if the constitution, or other governing document if there is no permanent constitution, has a section specifying how to replace the chief executive after a permanent vacancy. To aggregate the monthly data to a yearly level, I code succession rules as 1 if a majority of months have a succession rule and 0 otherwise.

Figure 1 graphs the proportion of years that each sub-Saharan African country has had succession rules during autocratic periods for the entire available dataset through 2020. Succession rules are common among African autocracies: over 77% of country-year observations have succession rules. Nearly half of countries, 21 of 47, have succession rules every year. Ethiopia is the only autocracy that never has succession rules, while Ghana and Lesotho are the only other autocracies that have succession rules in less than half of years. The commonality of succession rules underscores the core puzzle. Why autocracies have succession rules is not mysterious. Succession rules resolve essential problems in governing regimes. The more

interesting question is why a minority of autocracies lack succession rules.

Estimating the Probability of Coups

One of the main empirical challenges is measuring a dictator's risk of violent removal by elites within the regime. To measure the risk of violent removal, I use a Random Forest model to directly estimate the probability of a successful coup.³ Coups are identified using Powell and Thyne's (2011) data, updated through 2020. Powell and Thyne define coups as "illegal and overt attempts by the military or other elites within the state apparatus to unseat the sitting executive" (252; italics removed). Coups are successful if the perpetrators of the coup hold power for at least seven days. Powell and Thyne's definition captures the core concept in the argument: the threat of violent removal by actors already in the regime.

The model only predicts whether coups are successful, or whether the coup actually produces a turnover in leadership. Attempting a coup and succeeding in a coup have different data-generating processes; the difference between success and failure is not merely random (De Bruin 2017; Powell 2012). Dictators are most likely to fear the threat of coups when coups can credibly succeed. Indeed, strong dictators might even bait coups against them to suppress coup plotters and expand their hold on power (Luo and Rozenas 2023; Timoneda, Escribá-Folch, and Chin 2023). Failed coups, rather than frightening dictators, might present welcome opportunities for power concentration that dictators encourage.

The Random Forest predictions provide a more accurate, direct, and dynamic measurement of removal risk than previous strategies. A common strategy involves proxies such as export concentration or leader traits, but these proxies are static and rarely changing. The turnover rate, another common measurement, is more dynamic but decreases mechanically during a dictator's tenure. The Random Forest predictions, conversely, are dynamic and capture short-term fluctuations. Random Forest models also have advantages over generated regressors produced by standard regression models. Random Forest models do not impose functional forms on the predictors, allowing for larger and more complex sets of patterns than a researcher could identify. Although the inferential value of machine learning models like Random Forest is a de-

3. For detailed descriptions of Random Forest models, see Breiman (2001) and Muchlinski et al. (2016).

Table 1. Variables in the Random Forest Model for Predicting Coups

Variable	Source	Description/Notes
Assassinations	Banks and Wilson (2021)	Number of politically motivated murders or attempted murders of government officials and politics.
Cabinet Parties	Nyrup and Bramwell (2020)	Number of parties represented in the cabinet.
Civil Conflict Deaths	PRIO/UCDP*	Total battle-related deaths from civil conflict.
Coal Rents	WDI	Coal rents as a percentage of GDP.
Demonstrations	Banks and Wilson (2021)	Number of peaceful demonstrations with at least 100 people displaying opposition to government policies.
Excluded Pop.	EPR	Share of population in politically excluded ethnic groups (coded Powerless or Discriminated).
Gas Rents	WDI	Natural gas rents as a percentage of GDP.
GDP	PWT	Real gross domestic product in millions.
GDP Growth	PWT*	Annual percentage change in GDP.
Int. Conflict Deaths	PRIO/UCDP*	Total battle-related deaths from interstate conflict.
Leader Age	Archigos*	Age in years of chief executive. Averaged if multiple leaders.
Leader Tenure	Archigos*	Years since leader turnover.
Leg. Frac.	Banks and Wilson (2021)	Party fractionalization in the legislature using Rae's (1968) index.
Metal Rents	WDI	Mineral rents as a percentage of GDP.
Mil. Exp.	NMC	Military expenditure as state's total military budget.
Mil. Per.	NMC	Number of military personnel.
Oil Rents	WDI	Oil rents as a percentage of GDP.
Population	PWT	Total population.
Pop. Growth	PWT*	Annual percentage change in population.
Purges	Banks and Wilson (2021)	Number of systematic eliminations by jailing or executing political opposition.
Regime Age	Combined*	Number of consecutive years as a democracy or autocracy.
Riots	Banks and Wilson (2021)	Number of riots, or violent demonstrations of more than 100 citizens.
Strikes	Banks and Wilson (2021)	Number of strikes with 1,000 or more workers aimed at national government policies.
Terror	Banks and Wilson (2021)	Number of acts of terrorism or guerrilla warfare.
Total Coups	Powell and Thyne (2011)*	Total number of previous coups.
Years since Coup	Powell and Thyne (2011)*	Number of years since a successful coup occurred (or since 1946 if no coups have occurred).

Notes: * Author's calculations. Archigos version 4.1 (Goemans, Gleditsch, and Chiozza 2009); EPR refers to Ethnic Power Relations Core 2021 (Vogt et al. 2015); NMC refers to National Material Capabilities version 6 (Singer 1988; Singer, Bremer, and Stuckey 1972); PRIO/UCDP splices PRIO Battle Deaths 1966–1988 (Lacina and Gleditsch 2005) with UCDP Battle-Related Deaths v24.1 1989–2016 (Davies et al. 2024); Powell and Thyne (2011) data updated through 2020; PWT refers to Penn World Tables version 10.01 (Feenstra, Inklaar, and Timmer 2015); WDI refers to World Development Indicators. The datasets for regime classification are Boix, Miller, and Rosato (2012), Cheibub, Gandhi, and Vreeland (2010), and Geddes, Wright, and Frantz (2014).

veloping topic, they dominate regression models for pure prediction (Athey and Imbens 2019; Grimmer, Roberts, and Stewart 2021). Additionally, the predicted probability is estimated using only the subset of the model where that observation was not used during training, reducing the risk of overfitting.

The Random Forest model includes 26 variables as predictors, summarized in table 1. The choice of variables is based on existing forecasting models for coups and related forms of political instability.⁴ The variables were selected to balance predictive accuracy with minimizing the exclusion of cases due to missing data. The variables primarily focus on domestic political instability, economic conditions, political institutions, previous coups and regime changes, and resource wealth. To handle remaining missingness, I use multiple imputation with predictive mean matching.⁵ The sample includes all countries in sub-Saharan Africa during periods when they were autocracies from 1966 to 2016.⁶ The time period 1966 to 2016 is chosen to avoid excessive extrapolation based on the time periods included in the predictors.

Variables in Random Forest models lack direct interpretations. A variable can be associated with both an increase and decrease in coup probability depending on where the variable is split and earlier splits in the tree. Instead, variables can be ranked by importance according to Gini impurity, which measures how often variables are used to create predictions. Figure 2 displays the Gini impurity for each variable.⁷ The age of the regime is the most important variable, followed by a tier consisting of GDP growth, population, years since a coup, GDP, leader tenure, population growth, leader age, and military expenditure. International conflict deaths, conversely, is the least important variable.

Figure 3 evaluates model performance using a precision-recall curve. I use a precision-recall curve because coups are rare, so focusing on predicting coups is more important than simple classification. The standard summary statistic is the area under the curve (AUC). The baseline performance of a random classifier is the proportion of cases that experience coups, or

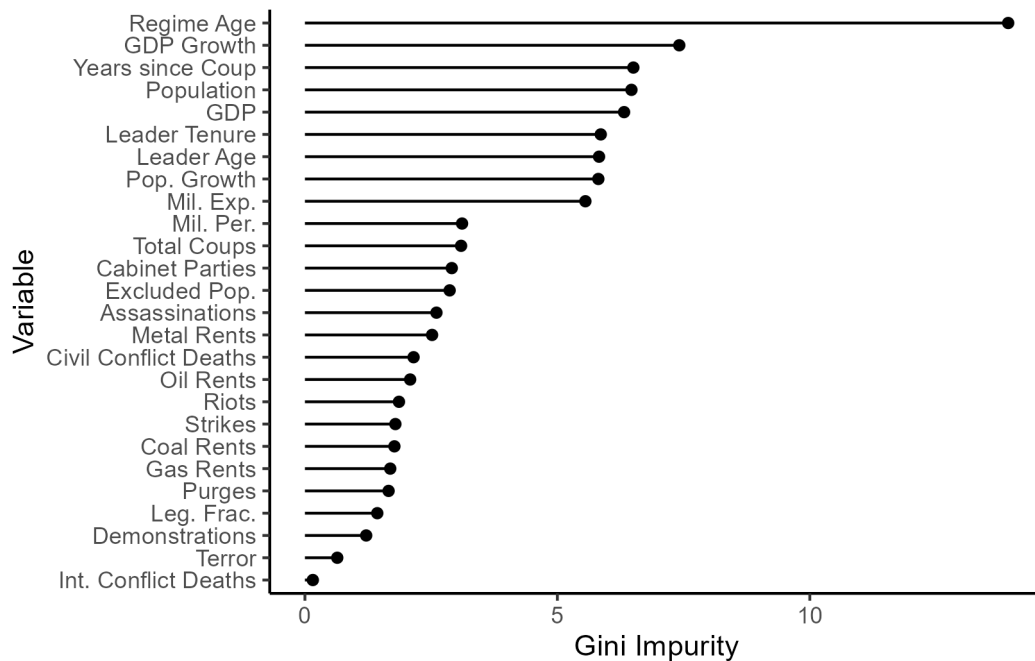
4. For example, see Belkin and Schofer (2003), Gassebner, Gutmann, and Voigt (2016), Gates et al. (2006), Goldstone et al. (2010), Londregan and Poole (1990), Miller, Joseph, and Ohl (2018), and Ward and Beger (2017).

5. Appendix B provides diagnostics on missing data imputation. The appendix shows that the results are robust to using only complete cases.

6. Table A1 lists the sample.

7. All point estimates in this section are calculated using Rubin's (1987) rules after multiple imputation with five imputed datasets.

Figure 2. Importance of Variables in the Random Forest Model for Predicting Coups



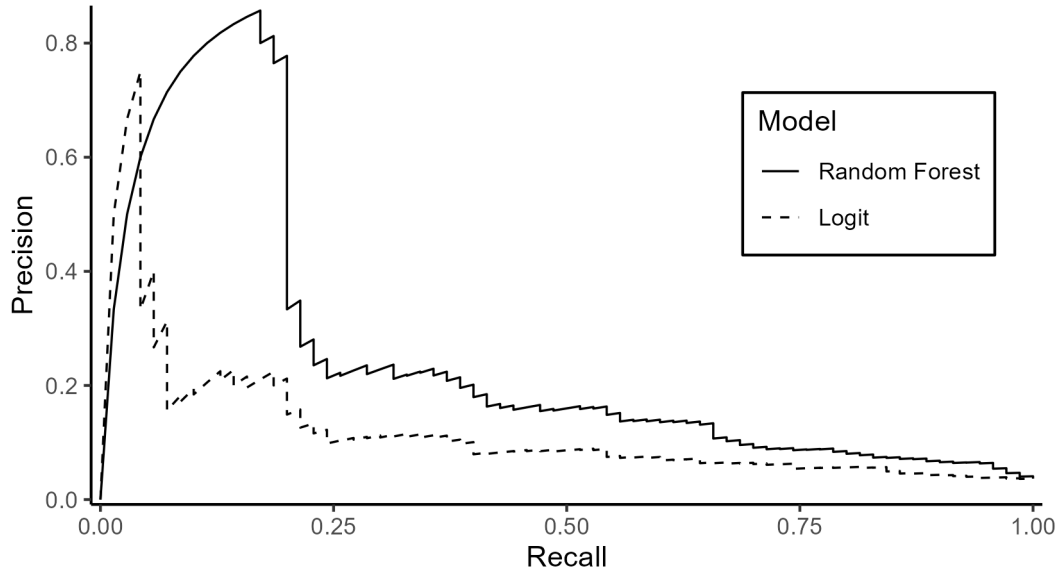
Notes: Variables are ranked according to Gini impurity. Variables with a higher reduction in Gini impurity are more important to the Random Forest model because they are used to split nodes more frequently.

0.0367. I also compare the Random Forest model to predictions from a simple logit model.⁸ The logit model has an AUC of 0.1137, which is over three times better than the random baseline. The Random Forest more than doubles the accuracy of the logit model with an AUC of 0.2543, nearly seven times better than the baseline. The Random Forest model performs substantially better than simpler alternatives.

Figure 4 demonstrates the importance of dynamic predictions. The figure graphs estimated coup probability for all 47 countries in the model during autocratic periods. There is no consistent pattern across countries. Some countries, such as Cameroon and Kenya, have consistently low probabilities, while others, such as Botswana and Eritrea, decline over time. Yet most countries experience large short-term fluctuations with no long-term patterns. For these countries, the predicted probabilities capture variation that existing proxies cannot.

8. To maintain comparability with the out-of-bag predictions from the Random Forest model, the logit predictions are produced using leave-one-out cross validation.

Figure 3. Precision Recall Curve for the Random Forest Model for Predicting Coups



Notes: Precision = $\frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$. Recall = $\frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$. The Random Forest has an area under the curve of 0.2543, 6.9 times better than a random classifier and 2.24 times better than the logit model.

Modeling Strategy

I model the relationship between coup probability and succession rules using probit models.

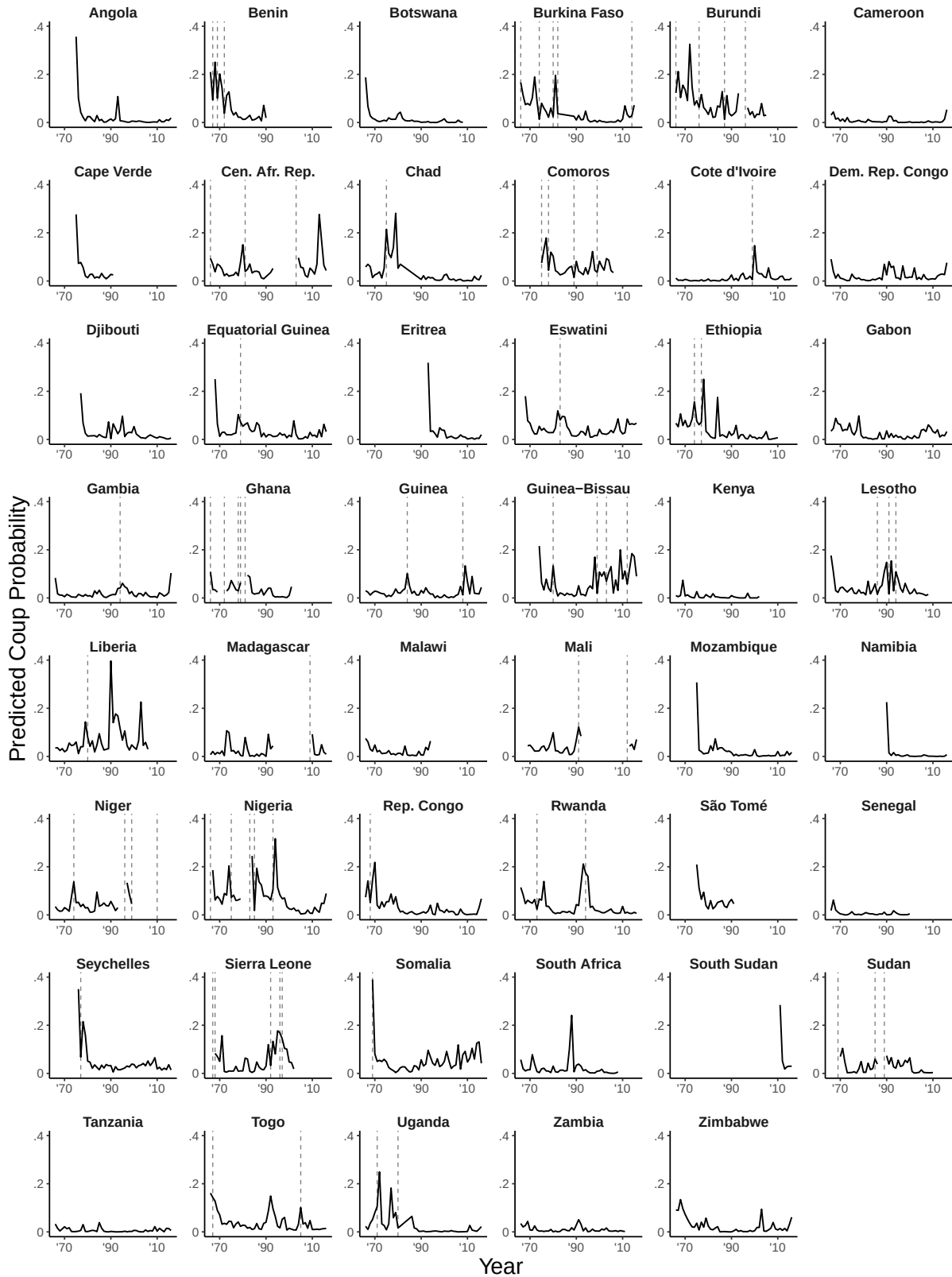
The probit model takes the form

$$\Pr(\text{Rule}_{it}) = \Phi(\beta \widehat{\text{Coup}}_{it-1} + \tau' Z_{it-1}), \quad (1)$$

where i indexes autocratic spells, which are consecutive years that a country was autocratic, and t indexes years. Rule_{it} is a dummy variable indicating the presence of succession rules. $\widehat{\text{Coup}}_{it-1}$ is the estimated probability of a coup based on the Random Forest model. Z_{it-1} is a vector of control variables. The independent variables are lagged one year to avoid simultaneity as the succession rule could change due to a coup. The key quantity of interest is the average marginal effect (AME) of coup probability on having a succession rule.

Handling uncertainty requires considerable care. In addition to standard sources, uncertainty is higher due to multiple imputation to handle missing data and variance in the Random Forest model. To address this uncertainty, I combine multiple imputation with bootstrapping (Hippel and Bartlett 2021; Schomaker and Heumann 2018). I use 1,000 bootstrap replications

Figure 4. Predicted Coup Probability by Country over Time



Notes: Predicted coup probability estimated by Random Forest. Vertical dashed lines represent years with at least one coup.

that resample autocratic spells with replacement. Within each bootstrapped sample, I impute a single round of predictive mean matching for missing data and train the Random Forest model. Then, I estimate the probit model and calculate the AME. The point estimate is the average of the bootstrapped samples, and I report uncertainty using percentile confidence intervals.

I use simple models that pool together introducing and retaining rules rather than focus on either one. This choice reflects both practical and theoretical considerations. Practically, the analysis already contends with considerable uncertainty. Separating introduction from retention further divides an already limited sample. Over 77% of country-years have succession rules, leaving few observations to estimate the introduction effect. A dynamic model that interacts every independent variable with the lagged dependent variable doubles the number of parameters while concentrating the introduction effect on a small and unrepresentative subset of the data. Pooling both effects into a single model preserves statistical power for the core test.

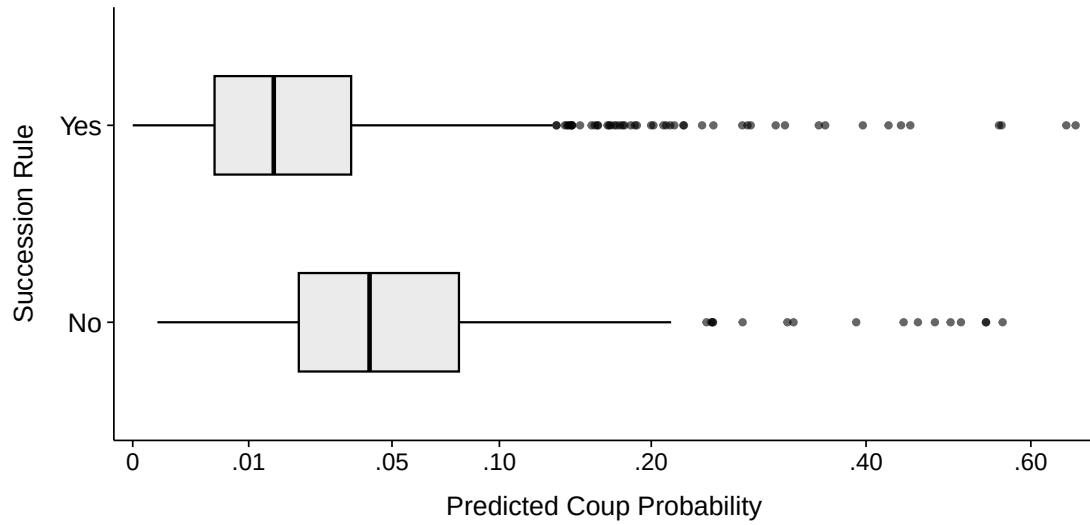
Theoretically, separating the two margins is not well motivated. Succession rules are not one-time choices that, once made, become permanent features of the regime. They are dynamic institutions that require ongoing maintenance. Dictators can and do remove succession rules when circumstances change. A dictator who retains a succession rule is making an active choice no less than a dictator who introduces one. The argument predicts that higher coup risk reduces the probability of having succession rules regardless of whether the regime had them previously. The pooled probit directly tests this prediction.

The models control for the predictors used in the Random Forest model. If any predictors also correlate with having succession rules, coup probability and succession rules could have a spurious correlation. Since the Random Forest model has 26 predictors that could be included in the regressions, the predictors are organized into eight models with one to five related controls.⁹ The models and their themes are:

1. Baseline: No controls.
2. Regime History: Age of regime, number of coups, and years since coup.
3. Elite management: Leader age, cabinet parties, party factionalization, and excluded pop-

9. Table A2 provides summary statistics for variables in the regression models.

Figure 5. Distribution of Coup Probability Based on Succession Rules



Notes: X-axis is square root transformed for presentation. The mean coup probability with succession rules is 3.07% and 6.91% without. The difference-in-means is significant at 99% ($t = 8.13$).

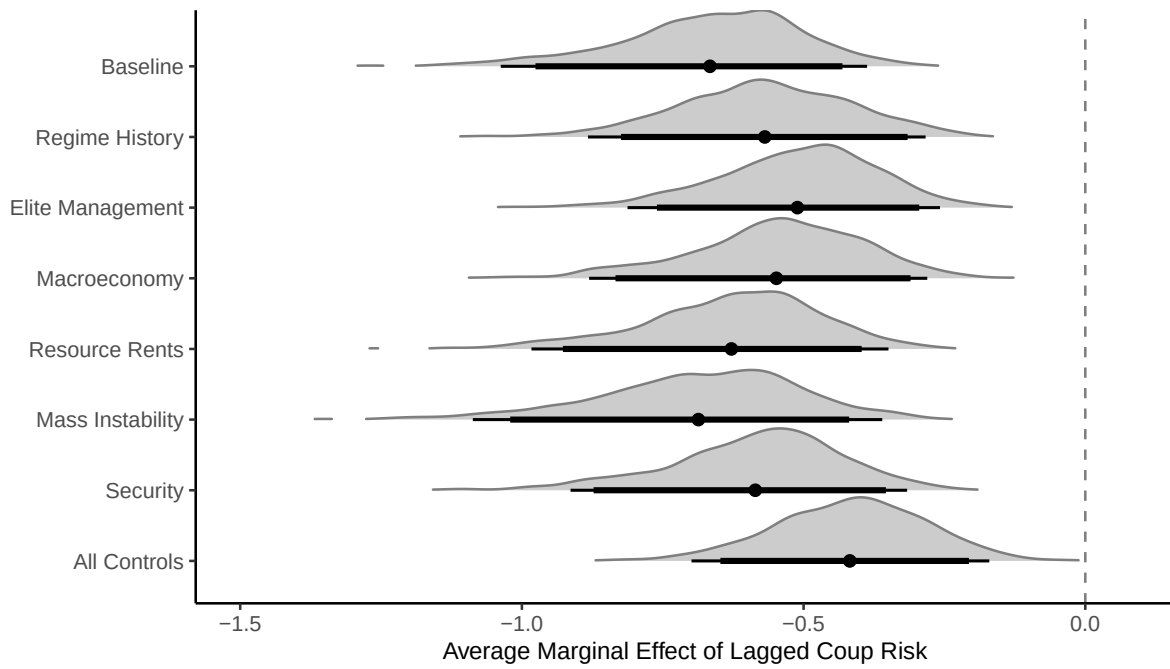
ulation share.

4. Macroeconomy: GDP, GDP growth, population, and population growth.
5. Resource rents: Wealth from oil, gas, coal, and metals.
6. Mass instability: Demonstrations, purges, riots, strikes, and terrorism.
7. Security: Military personnel, military expenditure, civil conflict deaths, and international conflict deaths.
8. Full: All predictors.

Results

Before moving to the regression results, figure 5 displays the distribution of coup probability based on whether a succession rule exists. Coup probability is substantially lower when succession rules exist, with a mean of 3.07% compared to 6.91%. The probability of a coup is more than twice as high in regimes without succession rules, and the difference-in-means is significant at 99% ($t = 8.13$). The distributions show further separation. The first quartile without

Figure 6. Average Marginal Effect of Coup Probability on Having Succession Rules



Notes: Distributions represent results from 1,000 clustered bootstrap samples. Thick lines represent 90% confidence intervals and thin lines 95% confidence intervals.

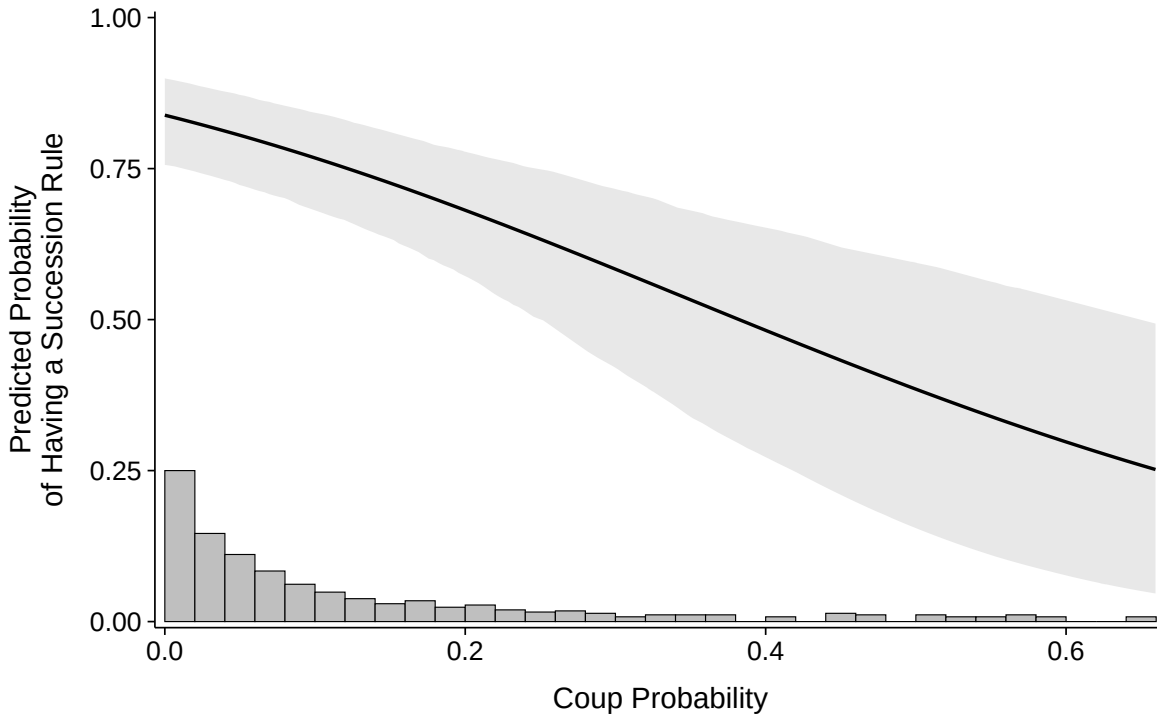
succession rules, 2.05%, is higher than the median with succession rules, 1.48%. Similarly, the median without succession rules, 4.17%, is higher than the third quartile with succession rules, 3.55%. Based on a simple bivariate analysis, coups are substantially more likely in regimes without succession rules.

Figure 6 shows the estimated AMEs of coup probability on having succession rules along with their bootstrapped distributions.¹⁰ The results support the prediction that autocracies are more likely to have succession rules when the probability of a coup is already low. All the point estimates are negative, as predicted, and none of the confidence intervals include 0. In fact, all eight bootstrapped distributions have only negative estimates with no bootstrapped sample producing a positive AME. Higher coup probability is significantly correlated with lower probabilities of having succession rules.

The strongest AME is the instability model. There, the AME is -0.687 , so each percentage point increase in coup probability is associated with a 0.687 percentage point decrease in the probability of having a succession rule. The 95% confidence interval in the instability model ranges from -1.087 to -0.33 . The weakest point estimate is the full model, but the AME is

10. Table A3 provides the full regression results.

Figure 7. Predicted Probability of Having Succession Rules



Notes: Grey area represents a 95% confidence interval based on 1,000 bootstrapped samples. The histogram at the bottom represents the distribution of predicted coup probability.

still substantial at -0.418 . The model with all controls also has the widest 95% confidence interval, which is unsurprising given the large number of parameters, at -0.699 to -0.17 . Even admitting that many of the predictors could be considered post-treatment and therefore attenuate the estimate, in addition to high multicollinearity, the negative association between coup probability and succession rules remains significant with meaningful effect sizes.

To contextualize the relationship, figure 7 shows the predicted probability of having a succession rule based on coup probability using the baseline model. When the predicted probability of a coup is near 0, the predicted probability of having a succession rule is 83.8%. At a coup probability of 10%, the predicted probability falls to 76.8%. At 20%, it declines to 68.2%, and at 30%, to 58.5%. The predicted probability crosses below 50% at a coup probability of around 40%. The decline is substantively meaningful across the range where most observations are concentrated. Most autocracies in the sample face coup probabilities below 10%, where the predicted probability of having a succession rule remains above 75%. But for the minority of regimes facing elevated coup risk, above around 15% to 20%, the model predicts a sharp

decline in the likelihood of having succession rules. The histogram at the bottom of the figure shows that observations thin out considerably above a coup probability of 10%, and the widening confidence interval reflects this sparsity rather than ambiguity regarding the relationship's direction.

Reverse Causality

Given existing arguments, the main inferential threat is reverse causality. If weak dictators adopt succession rules to prevent coups, then the correlation occurs because rules affect coup probability. I address the possibility of reverse causality with two methods, with results shown in Appendix C.

First, I estimate another Random Forest model with succession rules added as a predictor. Random Forest models are effective at identifying which variables among a large set have the most explanatory power (Arel-Bundock 2017). If succession rules are a weak predictor or do not add much predictive power, a spurious correlation is a less likely threat. Succession rules are, indeed, a weak predictor, ranking in the bottom third of importance alongside predictors such as strikes and coal rents. The AUC improves only barely: from 0.254 to 0.268. The predicted coup probabilities from the two models have an R^2 of 0.931, so there is little evidence that succession rules add much information. Given the weak predictive power of succession rules, they are unlikely to introduce a spurious correlation.

Second, I estimate a bivariate probit model. The bivariate probit model simultaneously estimates a model of coups at time t as a function of succession rules and controls at $t - 1$ with a model of succession rules at $t - 1$ as a function of coup probability and controls at $t - 2$. The full specification takes the form

$$\begin{aligned} \text{Coup}_{it}^* &= \alpha \text{Rule}_{it-1} + \gamma' Z_{it-1} + \varepsilon_1, \text{ Coup}_{it} = \mathbb{1}[\text{Coup}_{it}^* > 0], \\ \text{Rule}_{it-1}^* &= \beta \widehat{\text{Coup}}_{it-2} + \tau' Z_{it-2} + \varepsilon_2, \text{ Rule}_{it-1} = \mathbb{1}[\text{Rule}_{it-1}^* > 0], \\ \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \end{pmatrix} &\sim \mathcal{N} \left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \right]. \end{aligned} \tag{2}$$

The lag structure reflects the temporal ordering of the argument. The rule equation uses coup probability at $t - 2$ to predict rules at $t - 1$, mirroring the one-year lag in the main analysis. The coup equation uses rules at $t - 1$ to predict coups at t , testing whether rules affect subsequent coup outcomes. This structure ensures that the two dependent variables are measured at different time periods. The system, therefore, captures the sequential logic: coup risk shapes whether rules are adopted, and rules in turn may affect whether coups occur. By estimating both equations jointly, the bivariate probit accounts for shared unobservable determinants that could otherwise generate a spurious correlation.

The coefficient for coup probability in the rule equation is negative and significant in all eight models, as expected. The coefficient for succession rules in the coup equation is also negative and significant in seven models but not when all controls are included, which supports the existing argument that rules reduce coups and suggests that both arguments could exist concurrently. To ensure that this negative coefficient is not confounded, I rerun the models with $\widehat{\text{Coup}}_{it-1}$ included in the coup equation. Now, the coefficient flips to being positive in all eight models and is significant in five. The positive coefficient is consistent with my argument that succession rules can actually motivate coups. The bivariate probit models continue to support the hypothesis that higher coup probabilities are associated with lower probabilities of having succession rules and provide more tentative evidence that succession rules can otherwise spur coups. While reverse causality cannot be definitively ruled out, the amended Random Forest and bivariate probit models provide strong evidence that the correlation is not spurious.

Unobserved Heterogeneity

The probit model does not account for any time-invariant unobserved heterogeneity. Appendix D provides results from correlated random effects models, which are among the best-performing estimators for binary outcomes that are imbalanced (Crisman-Cox 2021). In all eight specifications with correlated random effects, the AME remains negative and significant, while the point estimates are, unsurprisingly, attenuated.

Coup Attempts

The main analysis uses only successful coups, but dictators may want to avoid any coup attempt. Appendix E replicates the entire analysis using coup attempt probability. Using coup attempts has a minimal effect on the results. The AME is negative and significant in all eight models.

Qualitative Evidence

Directly observing the dynamics of autocratic succession is challenging due to the secrecy of autocratic regimes, particularly regarding high-level politics. Zimbabwe is an exception due to extensive biographies of long-time dictator Robert Mugabe and a more public constitution writing process in 2013. Appendix F briefly describes succession in Zimbabwe with two conclusions. First, Mugabe adopted a formal succession rule after he consolidated power, consistent with this paper's argument. Second, the ruling party weakened the constitutional successor's position after Mugabe's power declined due to fears that the vice-president could stage a coup.

Conclusion

Autocracies vary in their willingness and ability to regulate political succession. A recent wave of research asks why some autocracies formalize succession in the constitution. According to this research, dictators use succession rules, particularly rules involving designated successors, to reduce the threat of being overthrown. In contrast, I argue that dictators have succession rules when the probability of a coup is already low. Succession rules address a fundamental task of governance by preparing the regime for a leader's demise. Succession rules, however, can also create threats to the dictator's rule by empowering a successor and angering groups of elites. Given the threats created by succession rules, dictators are most likely to have them when they are confident in their ability to suppress these threats.

I test this argument using original data on autocratic succession rules in sub-Saharan Africa from 1966 to 2016 and a new approach for measuring a dictator's security. I use Random

Forest, a machine learning algorithm, to predict the probability that a successful coup occurs. Based on the regression models, higher predicted probabilities of a coup are associated with lower probabilities of having succession rules.

Future research should explore the dynamics of informal succession. Dictators can informally designate successors without formal succession rules, and informal succession plans can even supplant formal rules. In the case of Zimbabwe (discussed in Appendix F), for instance, Robert Mugabe typically balanced a successor from the formal line of succession against a potential successor outside it—initially First Vice-President Joice Mujuru against Minister of Defence Emmerson Mnangagwa, then Mnangagwa as first vice-president against First Lady Grace Mugabe. Identifying informal succession processes poses significant challenges. By definition, informal processes are not written down, and if the plan is never implemented, scholars may never know that it existed. Nonetheless, studying the informal dynamics of succession in autocracies is necessary for a complete understanding of autocratic succession.

Future research could also examine the role of succession outside domestic politics. Existing studies, including this one, primarily focus on the domestic politics of how succession rules emerge and their impact on political survival. Succession matters to anybody interested in a country given succession's role in influencing a regime's stability and policies. International actors are also likely to care about an autocracy's succession processes. Foreign investors and bondholders may be particularly attuned to succession, as it could both secure and endanger their investments. Given the argument and findings of this paper, succession rules, ultimately, signal stability and strength, not constraints.

Finally, the most direct implication of my findings is that, since they contradict the expectations of existing literature, the effects of succession rules on autocratic survival need revisiting. Existing research finds a positive relationship between succession rules and a dictator's political survival; however, these findings may be spurious if dictators have succession rules when they are already likely to survive. Regardless of whether the relationship persists, future research should consider additional motivations for having successors. The variation in how autocracies handle succession is vast. New theories and new data are needed to better explain variation in autocratic succession.

The conclusions of this paper ultimately speak to the fundamental challenge of studying institutions described in the Riker Objection. Institutions are not exogenous forces. Instead, they are political outcomes themselves, often shaped by the same forces that influence the outcomes they are meant to explain (Pepinsky 2014; Riker 1980). The comparative literature on autocracy and institutions rarely studies institutions as outcomes. Scholars who account for potential endogeneity typically treat the causes of institutions as either a statistical nuisance or a design problem. However, neither approach is sufficient to address endogeneity comprehensively. Institutions have complex causes that are difficult to address as objects of secondary interest, especially in a cross-national context. Rather than treating endogeneity as merely a problem, scholars should study institutions as outcomes in their own right and subsequently evaluate their effects. To understand the effects of institutions, we need to understand their causes.

Competing Interests None.

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References

- Albertus, Michael, and Victor Menaldo. 2012. "Dictators as Founding Fathers? The Role of Constitutions under Autocracy." *Economics & Politics* 24 (3): 279–306.
- Arel-Bundock, Vincent. 2017. "The Political Determinants of Foreign Direct Investment: A Firm-Level Analysis." *International Interactions* 43 (3): 424–52.
- Arriola, Leonardo R. 2009. "Patronage and Political Stability in Africa." *Comparative Political Studies* 42 (10): 1339–62.
- Athey, Susan, and Guido W. Imbens. 2019. "Machine Learning Methods That Economists Should Know About." *Annual Review of Economics* 11 (August): 685–725.
- Banks, Arthur S., and Kenneth A. Wilson. 2021. "Cross-National Time-Series Data (CNTS)." Jerusalem, Israel: Databank International. <https://www.cntsdata.com/>.
- Barros, Robert. 2002. *Constitutionalism and Dictatorship: Pinochet, the Junta, and the 1980 Constitution*. Cambridge: Cambridge University Press.

- Belkin, Aaron, and Evan Schofer. 2003. "Toward a Structural Understanding of Coup Risk." *Journal of Conflict Resolution* 47 (5): 594–620.
- Boix, Carles, Michael Miller, and Sebastian Rosato. 2012. "A Complete Data Set of Political Regimes, 1800–2007." *Comparative Political Studies* 46 (12): 1523–54.
- Boix, Carles, and Milan W. Svolik. 2013. "The Foundations of Limited Authoritarian Government: Institutions, Commitment, and Power-Sharing in Dictatorships." *Journal of Politics* 75 (2): 300–16.
- Boucoyannis, Deborah. 2021. *Kings as Judges: Power, Justice, and the Origins of Parliaments*. Cambridge: Cambridge University Press.
- Breiman, Leo. 2001. "Random Forests." *Machine Learning* 45 (1): 5–32.
- Brownlee, Jason. 2007. *Authoritarianism in an Age of Democratization*. Cambridge: Cambridge University Press.
- Carey, John M. 2000. "Parchment, Equilibria, and Institutions." *Comparative Political Studies* 33 (6/7): 735–61.
- Cheibub, José Antonio, Jennifer Gandhi, and James Raymond Vreeland. 2010. "Democracy and Dictatorship Revisited." *Public Choice* 143 (1–2): 67–101.
- Crisman-Cox, Casey. 2021. "Estimating Substantive Effects in Binary Outcome Panel Models: A Comparison." *Journal of Politics* 83 (2): 532–46.
- Davies, Shawn, Therése Pettersson, Margareta Sollenberg, and Magnus Öberg. 2024. "Organized Violence 1989–2024, and the Challenges of Identifying Civilian Victims." *Journal of Peace Research* 62 (4): 1223–40.
- De Bruin, Erica. 2017. "Preventing Coups d'état: How Counterbalancing Works." *Journal of Conflict Resolution* 62 (7): 1433–58.
- Elkins, Zachary, Tom Ginsburg, and James Melton. 2014. "Characteristics of National Constitutions, Version 2.0." Comparative Constitutions Project. <https://comparativeconstitutionsproject.org>.
- Feenstra, Robert C., Robert Inklaar, and Marcel P. Timmer. 2015. "The Next Generation of the Penn World Table." *American Economic Review* 105 (10): 3150–82.
- Francois, Patrick, Ilia Rainer, and Francesco Trebbi. 2015. "How Is Power Shared in Africa?" *Econometrica* 83 (2): 465–503.
- Frantz, Erica, and Elizabeth A. Stein. 2017. "Countering Coups: Leadership Succession Rules in Dictatorships." *Comparative Political Studies* 50 (7): 935–62.
- Gandhi, Jennifer. 2008. *Political Institutions under Dictatorship*. Cambridge: Cambridge University Press.
- Gandhi, Jennifer, and Adam Przeworski. 2007. "Authoritarian Institutions and the Survival of Autocrats." *Comparative Political Studies* 40 (11): 1279–301.
- Gassebner, Martin, Jerg Gutmann, and Stefan Voigt. 2016. "When to Expect a Coup d'état? An Extreme Bounds Analysis of Coup Determinants." *Public Choice* 169 (3–4): 293–313.

- Gates, Scott, Håvard Hegre, Mark P. Jones, and Håvard Strand. 2006. "Institutional Inconsistency and Political Instability: Polity Duration, 1800–2000." *American Journal of Political Science* 50 (4): 893–908.
- Geddes, Barbara. 1999. "What Do We Know about Democratization after Twenty Years?" *Annual Review of Political Science* 2 (June): 115–44.
- Geddes, Barbara, Joseph Wright, and Erica Frantz. 2014. "Autocratic Breakdown and Regime Transitions: A New Data Set." *Perspectives on Politics* 12 (2): 313–31.
- Geddes, Barbara, Joseph Wright, and Erica Frantz. 2018. *How Dictatorships Work: Power, Personalization, and Collapse*. Cambridge: Cambridge University Press.
- Goemans, Henk E., Kristian Skrede Gleditsch, and Giacomo Chiozza. 2009. "Introducing Archigos: A Dataset of Political Leaders." *Journal of Peace Research* 46 (2): 269–83.
- Goldstone, Jack A., Robert H. Bates, David L. Epstein, Ted Robert Gurr, Michael B. Lustik, Monty G. Marshall, Jay Ulfelder, and Mark Woodward. 2010. "A Global Model for Forecasting Political Instability." *American Journal of Political Science* 54 (1): 190–208.
- Grimmer, Justin, Margaret E. Roberts, and Brandon M. Stewart. 2021. "Machine Learning for Social Science: An Agnostic Approach." *Annual Review of Political Science* 24 (May): 395–419.
- Hardin, Russell. 1989. "Why a Constitution?" In *The Federalist Papers and the New Institutionalism*, edited by Bernard Grofman and Donald Wittman, 100–120. New York: Agathon.
- Herz, John H. 1952. "The Problems of Successorship in Dictatorial Régimes: A Study in Comparative Law and Institutions." *Journal of Politics* 14 (1): 19–40.
- Hippel, Paul T. von, and Jonathan W. Bartlett. 2021. "Maximum Likelihood Multiple Imputation: Faster Imputations and Consistent Standard Errors Without Posterior Draws." *Statistical Science* 36 (3): 400–420.
- Kokkonen, Andrej, Jørgen Møller, and Anders Sundell. 2022. *The Politics of Succession: Forging Stable Monarchies in Europe, AD 1000–1800*. Oxford: Oxford University Press.
- Kokkonen, Andrej, and Anders Sundell. 2014. "Delivering Stability—Primogeniture and Autocratic Survival in European Monarchies 1000–1800." *American Political Science Review* 108 (2): 438–53.
- Konrad, Kai A., and Vai-Lam Mui. 2017. "The Prince—or Better No Prince? The Strategic Value of Appointing a Successor." *Journal of Conflict Resolution* 61 (10): 2158–82.
- Kurrild-Klitgaard, Peter. 2000. "The Constitutional Economics of Autocratic Succession." *Public Choice* 103 (1–2): 63–84.
- Lacina, Bethany, and Nils Petter Gleditsch. 2005. "Monitoring Trends in Global Combat: A New Dataset of Battle Deaths." *European Journal of Population* 21 (2–3): 145–66.
- Levi, Margaret. 1988. *Of Rule and Revenue*. Berkeley: University of California Press.
- Londregan, John B., and Keith T. Poole. 1990. "Poverty, the Coup Trap, and the Seizure of Executive Power." *World Politics* 42 (2): 151–83.

- Luo, Zhaotian, and Arturas Rozenas. 2023. "Ruling the Ruling Coalition: Information Control and Authoritarian Power-Sharing." *Quarterly Journal of Political Science* 18 (2): 183–213.
- Magaloni, Beatriz. 2008. "Credible Power-Sharing and the Longevity of Authoritarian Rule." *Comparative Political Studies* 41 (4/5): 715–41.
- Mattes, Michaela, and Mariana Rodríguez. 2014. "Autocracies and International Cooperation." *International Studies Quarterly* 58 (3): 527–38.
- Meng, Anne. 2020. *Constraining Dictatorship: From Personalized Rule to Institutionalized Regimes*. Cambridge: Cambridge University Press.
- Meng, Anne. 2021. "Winning the Game of Thrones: Leadership Succession in Modern Autocracies." *Journal of Conflict Resolution* 65 (5): 950–81.
- Meng, Anne, and Jack Paine. 2022. "Power Sharing and Authoritarian Stability: How Rebel Regimes Solve the Guardianship Dilemma." *American Political Science Review* 116 (4): 1208–55.
- Meng, Anne, Jack Paine, and Robert Powell. 2023. "Authoritarian Power Sharing: Concepts, Mechanisms, and Strategies." *Annual Review of Political Science* 26 (June): 153–73.
- Miller, Michael K., Michael Joseph, and Dorothy Ohl. 2018. "Are Coups Really Contagious? An Extreme Bounds Analysis of Political Diffusion." *Journal of Conflict Resolution* 62 (2): 410–41.
- Muchlinski, David, David Siroky, Jingrui He, and Matthew Kocher. 2016. "Comparing Random Forest with Logistic Regression for Predicting Class-Imbalanced Civil War Onset Data." *Political Analysis* 24 (1): 87–103.
- Myerson, Roger B. 2008. "The Autocrat's Credibility Problem and Foundations of the Constitutional State." *American Political Science Review* 102 (1): 125–39.
- Nyrup, Jacob, and Stuart Bramwell. 2020. "Who Governs? A New Global Dataset on Members of Cabinets." *American Political Science Review* 114 (4): 1366–74.
- Pepinsky, Thomas. 2014. "The Institutional Turn in Comparative Authoritarianism." *British Journal of Political Science* 44 (3): 631–53.
- Powell, Jonathan. 2012. "Determinants of the Attempting and Outcome of Coups d'état." *Journal of Conflict Resolution* 56 (6): 1017–40.
- Powell, Jonathan N., and Clayton L. Thyne. 2011. "Global Instances of Coups from 1950 to 2010: A New Dataset." *Journal of Peace Research* 48 (2): 249–59.
- Przeworski, Adam. 2013. "Ruling against Rules." In *Constitutions in Authoritarian Regimes*, edited by Tom Ginsburg and Alberto Simpser, 21–35. Cambridge: Cambridge University Press.
- Rae, Douglas. 1968. "A Note on the Fractionalization of Some European Party Systems." *Comparative Political Studies* 1 (3): 413–18.
- Riker, William H. 1980. "Implications from the Disequilibrium of Majority Rule for the Study of Institutions." *American Political Science Review* 74 (2): 432–46.

- Rubin, Donald B. 1987. *Multiple Imputation for Nonresponse in Surveys*. New York: John Wiley & Sons.
- Schomaker, Michael, and Christian Heumann. 2018. "Bootstrap Inference When Using Multiple Imputation." *Statistics in Medicine* 37 (14): 2252–66.
- Singer, J. David. 1988. "Reconstructing the Correlates of War Dataset on Material Capabilities of States, 1816–1985." *International Interactions* 14 (2): 115–32.
- Singer, J. David, Stuart Bremer, and John Stuckey. 1972. "Capability Distribution, Uncertainty, and Major Power War, 1820–1965." In *Peace, War, and Numbers*, edited by Bruce Russett, 19–48. Beverly Hills, CA: Sage.
- Slater, Dan. 2003. "Iron Cage in an Iron Fist: Authoritarian Institutions and the Personalization of Power in Malaysia." *Comparative Politics* 36 (1): 81–101.
- Svolik, Milan W. 2012. *The Politics of Authoritarian Rule*. Cambridge: Cambridge University Press.
- Timoneda, Joan C., Abel Escribá-Folch, and John Chin. 2023. "The Rush to Personalize: Power Concentration after Failed Coups in Dictatorships." *British Journal of Political Science* 53 (3): 878–901.
- Tullock, Gordon. 1987. *Autocracy*. Dordrecht, The Netherlands: Springer-Science+Business Media.
- Vogt, Manuel, Nils-Christian Bormann, Seraina Rüegger, Lars-Erik Cederman, Philipp Hunziker, and Luc Girardin. 2015. "Integrating Data on Ethnicity, Geography, and Conflict: The Ethnic Power Relations Data Set Family." *Journal of Conflict Resolution* 59 (7): 1327–42.
- Ward, Michael D., and Andreas Beger. 2017. "Lessons from Near Real-Time Forecasting of Irregular Leadership Changes." *Journal of Peace Research* 54 (2): 141–56.
- Weeks, Jessica L. 2012. "Strongmen and Straw Men: Authoritarian Regimes and the Initiation of International Conflict." *American Political Science Review* 106 (2): 326–47.
- Williamson, Scott. 2024. *The King Can Do No Wrong: Blame Games and Power Sharing in Authoritarian Regimes*. Cambridge: Cambridge University Press.
- Wright, Joseph. 2008. "Do Authoritarian Institutions Constrain? How Legislatures Affect Economic Growth and Investment." *American Journal of Political Science* 52 (2): 322–43.
- Zhou, Congyi. 2023. "Last Step to the Throne: The Conflict between Rulers and Their Successors." *Political Science Research & Methods* 11 (1): 80–94.

Supplementary Material

Succession from Strength: Coups and Autocratic Succession Rules in Sub-Saharan Africa

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A Additional Tables

Table A1. Sample

Country	Years	Country	Years
Angola	1975–2016	Liberia	1966–2006
Benin	1966–1990	Madagascar	1966–1993, 2009–2016
Botswana	1966–2008	Malawi	1966–1994
Burkina Faso	1966–1982, 1989–2015	Mali	1969–1992, 2012–2016
Burundi	1966–1993, 1996–2005, 2015–2016	Mozambique	1975–2016
Cameroon	1966–2016	Namibia	1990–2016
Cape Verde	1975–1991	Niger	1966–1993, 1996–1999, 2010–2011, 2016
Central African Republic	1966–1993, 2003–2016	Nigeria	1966–1979, 1983–2016
Chad	1966–1981, 1990–2016	Republic of the Congo	1966–2016
Comoros	1975–2006	Rwanda	1966–2016
Côte d'Ivoire	1966–2016	São Tomé & Príncipe	1975–1991
Democratic Republic of the Congo	1966–2016	Senegal	1966–2000
Djibouti	1977–2016	Seychelles	1976–2016
Equatorial Guinea	1968–2016	Sierra Leone	1967–2002
Eritrea	1993–2016	Somalia	1969–2016
Eswatini	1968–2016	South Africa	1966–2008
Ethiopia	1966–2016	South Sudan	2011–2016
Gabon	1966–2016	Sudan	1969–1986, 1989–2016
Gambia	1966–2016	Tanzania	1966–2016
Ghana	1966–1969, 1972–1979, 1981–2001	Togo	1966–2016
Guinea	1966–2016	Uganda	1966–1980, 1986–2016
Guinea-Bissau	1974–2016	Zambia	1966–2011
Kenya	1966–2002	Zimbabwe	1966–2016
Lesotho	1966–2008		

Table A2. Summary Statistics

Variable	<i>N</i>	Mean	SD	Min	Q1	Median	Q3	Max
Succession Rule	1909	0.808	0.394	0.000	1.000	1.000	1.000	1.000
Successful Coup	1909	0.037	0.188	0.000	0.000	0.000	0.000	1.000
Leader Age	1798	55.743	11.932	19.000	47.000	55.000	64.000	91.000
Assassinations	1808	0.085	0.456	0.000	0.000	0.000	0.000	8.000
Cabinet Parties	1835	1.799	1.941	0.000	1.000	1.000	2.000	17.000
Logged Civil Deaths	1909	1.243	2.586	0.000	0.000	0.000	0.000	10.814
Coal Rents	1423	0.361	3.649	0.000	0.000	0.000	0.000	69.804
Total Coups	1909	1.211	1.566	0.000	0.000	1.000	2.000	7.000
Years since Coup	1909	18.615	19.468	0.000	6.000	14.000	26.000	133.000
Regime Age	1897	28.690	28.517	0.000	11.000	22.000	36.000	159.000
Demonstrations	1808	0.264	1.109	0.000	0.000	0.000	0.000	26.000
Excluded Pop.	1851	0.238	0.293	0.000	0.000	0.100	0.440	0.980
Logged GDP	1828	9.299	1.533	5.531	8.243	9.315	10.310	13.789
GDP Growth	1826	0.039	0.077	-0.510	0.007	0.038	0.067	1.063
Gas Rents	1372	0.067	0.344	0.000	0.000	0.000	0.000	3.968
Logged Int. Deaths	1909	0.050	0.565	0.000	0.000	0.000	0.000	10.820
Leg. Frac.	1501	0.159	0.241	0.000	0.000	0.000	0.315	0.968
Metal Rents	1617	0.945	2.814	0.000	0.000	0.005	0.320	36.853
Logged Mil. Exp.	1706	10.562	2.564	0.000	9.679	10.858	11.958	15.739
Mil. Per.	1855	28.519	47.048	0.000	4.000	10.000	30.000	352.000
Oil Rents	1435	3.909	10.182	0.000	0.000	0.000	0.380	82.777
Logged Population	1828	1.575	1.574	-2.796	0.505	1.795	2.646	5.226
Pop. Growth	1826	0.026	0.011	-0.070	0.022	0.027	0.031	0.104
Purges	1808	0.074	0.372	0.000	0.000	0.000	0.000	5.000
Riots	1808	0.291	1.154	0.000	0.000	0.000	0.000	23.000
Strikes	1808	0.042	0.273	0.000	0.000	0.000	0.000	4.000
Terror	1808	0.465	4.145	0.000	0.000	0.000	0.000	121.000
Leader Tenure	1800	9.485	8.365	0.000	3.000	7.000	14.000	41.000

Table A3. Regression Results for Probit Models of Succession Rules

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Coup Risk	-2.616*** (0.698)	-2.463*** (0.672)	-2.319*** (0.655)	-2.340*** (0.699)	-2.545*** (0.702)	-2.746*** (0.798)	-2.501*** (0.688)	-2.414*** (0.798)
Regime Age		-0.006 (0.011)						-0.004 (0.010)
Total Coups		-0.043 (0.063)						-0.097 (0.086)
Years since Coup		0.026*** (0.012)						0.018*** (0.015)
Leader Age			0.024*** (0.010)					0.010 (0.015)
Cabinet Parties			0.264*** (0.130)					0.334*** (0.162)
Leg. Frac.			0.168 (0.456)					0.160 (0.639)
Excluded Pop.			-0.768 (0.414)					-0.803*** (0.401)
Logged GDP				0.429*** (0.151)				0.391*** (0.202)
GDP Growth				0.557 (0.704)				0.708 (0.896)
Logged Population				-0.431*** (0.162)				-0.359*** (0.174)
Pop. Growth				-3.028 (10.889)				-3.105 (12.739)
Oil Rents					0.014 (0.066)			-0.003 (0.036)
Gas Rents					1.726 (2.804)			1.230 (2.685)
Coal Rents					0.659 (3.629)			0.482 (3.515)
Metal Rents					0.087 (0.081)			0.081 (0.097)
Demonstrations						0.175*** (0.098)		0.121 (0.112)
Purges						-0.257 (0.131)		-0.197 (0.139)
Riots						0.007 (0.090)		-0.004 (0.115)
Strikes						0.227 (0.334)		0.254 (0.378)
Terror						0.064 (0.074)		0.188*** (0.127)
Logged Mil. Exp.							0.100*** (0.041)	0.066 (0.059)
Mil. Per.							-0.006 (0.003)	-0.005 (0.004)
Logged Civil Deaths							-0.044 (0.034)	-0.073 (0.038)
Logged Int. Deaths							0.056 (0.165)	0.086 (0.170)
Intercept	0.999*** (0.151)	0.873*** (0.352)	-0.498 (0.538)	-2.165 (1.288)	0.870*** (0.168)	0.976*** (0.152)	0.207 (0.471)	-3.238*** (1.500)
N	1862	1862	1862	1862	1862	1862	1862	1862

Notes: *** 95% confidence interval based on 1,000 clustered bootstrapped samples excludes 0.
Standard errors based on 1,000 clustered bootstrapped samples in parentheses.
All independent variables lagged by one year.

B Missing Data

Table B1. Missing Data in Random Forest Predictors

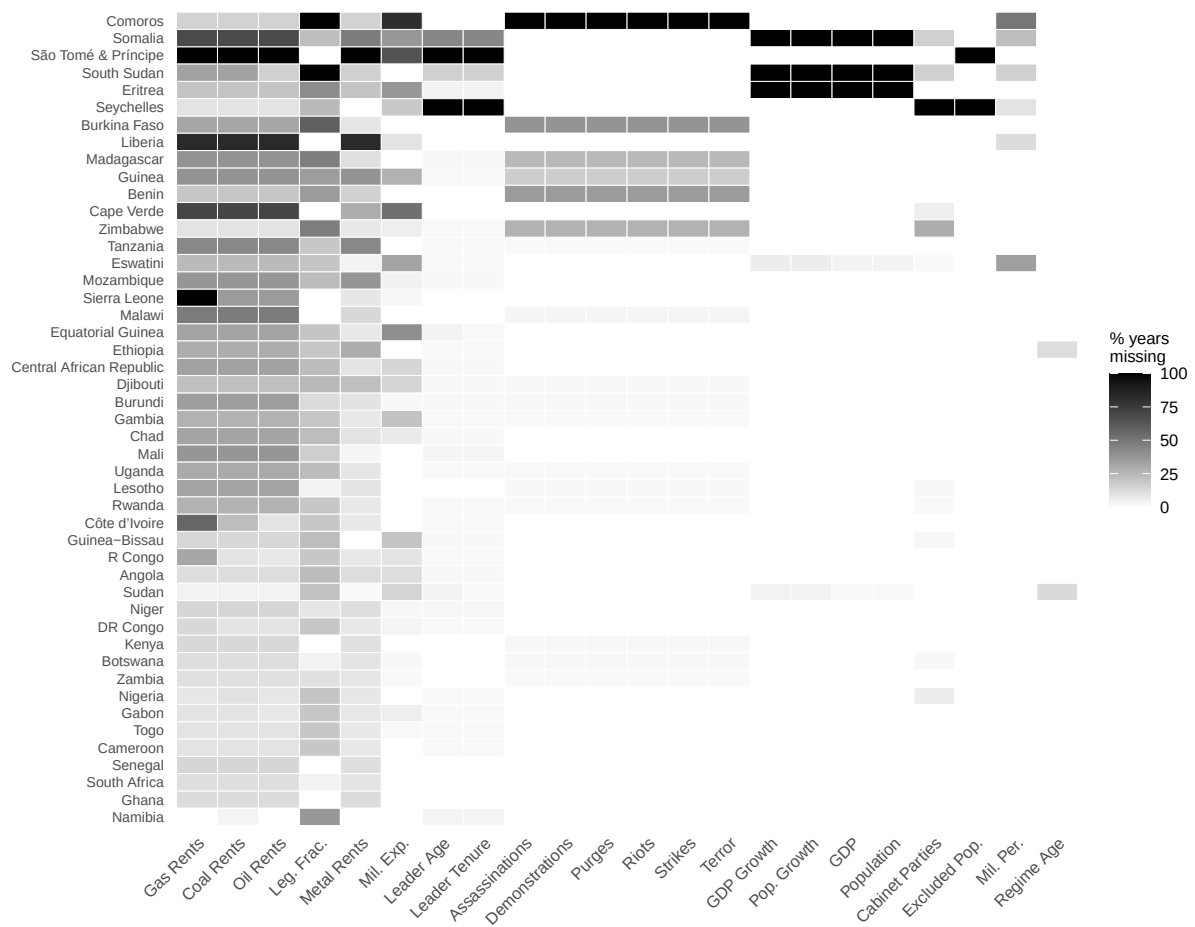
Variable	Observed	Missing	% Missing
Gas Rents	1372	537	28.1
Coal Rents	1423	486	25.5
Oil Rents	1435	474	24.8
Leg. Frac.	1501	408	21.4
Metal Rents	1617	292	15.3
Mil. Exp.	1706	203	10.6
Leader Age	1798	111	5.8
Leader Tenure	1800	109	5.7
Assassinations	1808	101	5.3
Demonstrations	1808	101	5.3
Purges	1808	101	5.3
Riots	1808	101	5.3
Strikes	1808	101	5.3
Terror	1808	101	5.3
GDP Growth	1826	83	4.3
Pop. Growth	1826	83	4.3
GDP	1828	81	4.2
Population	1828	81	4.2
Cabinet Parties	1835	74	3.9
Excluded Pop.	1851	58	3.0
Mil. Per.	1855	54	2.8
Regime Age	1897	12	0.6
Civil Conflict Deaths	1909	0	0.0
Total Coups	1909	0	0.0
Years since Coup	1909	0	0.0
Int. Conflict Deaths	1909	0	0.0
Total	45882	3752	7.6

Notes: Sample: 1909 country-years, 47 sub-Saharan African autocracies, 1966–2016.
Rows sorted by percentage missing (descending).

Table B1 summarizes the extent of missing data across the 26 Random Forest predictor variables in the sample. Of the 26 predictors, 22 have at least some missing data. The most-missing variables are the resource rent measures—gas rents (28.1%), coal rents (25.5%), oil rents (24.8%)—and legislative fractionalization (21.4%), reflecting the limited early-period coverage of the WDI and Banks data sources. Figure B1 displays the pattern of missingness across countries and variables, showing whether missingness is concentrated in particular countries or time periods.

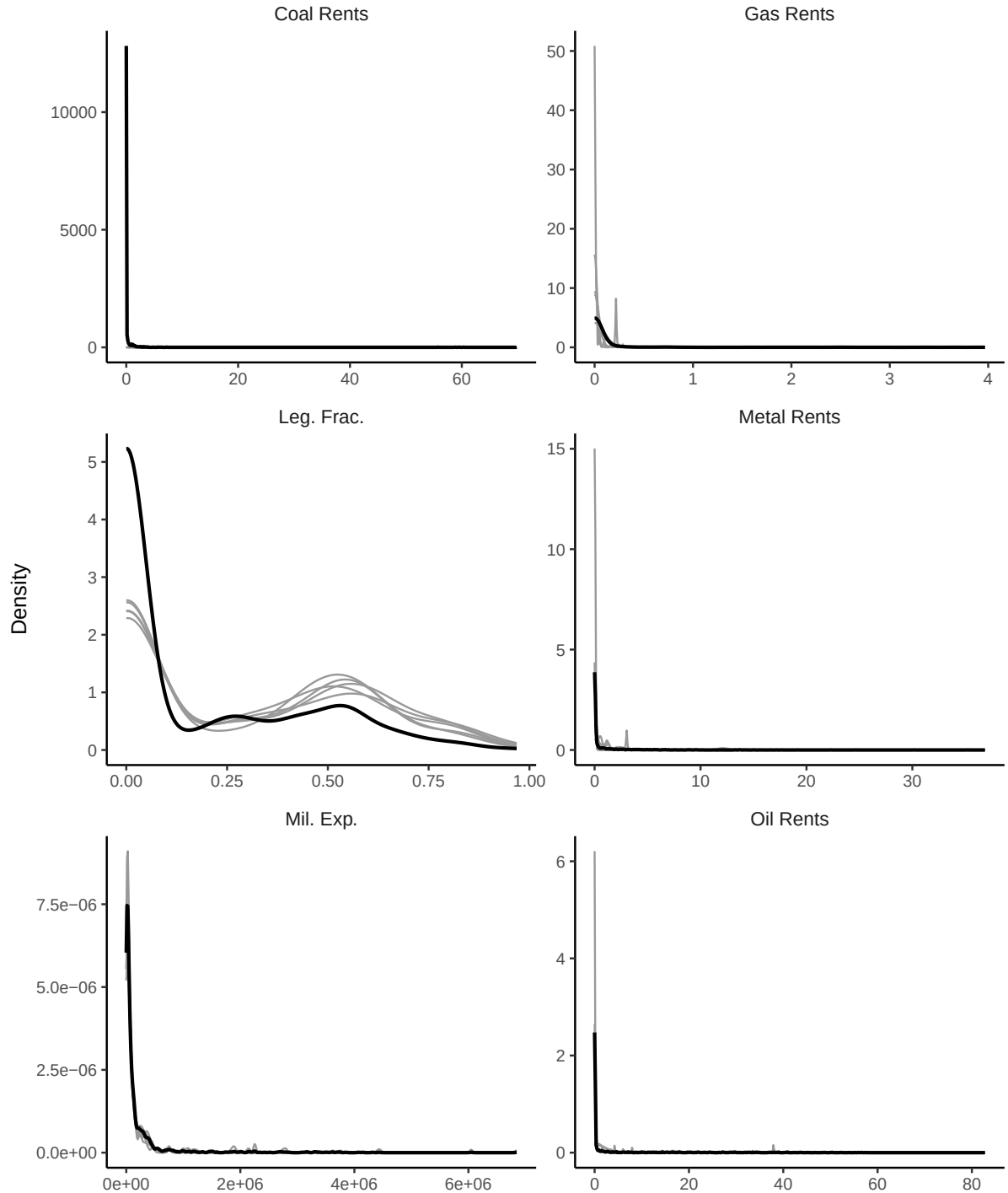
Figure B2 compares the distribution of observed values to the distribution of imputed values for the variables with the highest rates of missingness. The imputed distributions closely track the observed distributions, providing no evidence of implausible imputed values. Table B2 confirms this numerically: the means and standard deviations of imputed values are similar to those of observed values for all high-missingness variables.

Figure B1. Pattern of Missing Data across Countries and Predictor Variables



Notes: Cell shading indicates the percentage of country-years with missing data for each country–variable combination. Variables are sorted by overall percentage missing (left to right) and countries by total missingness (bottom to top).

Figure B2. Observed vs. Imputed Distributions for Most-Missing Variables



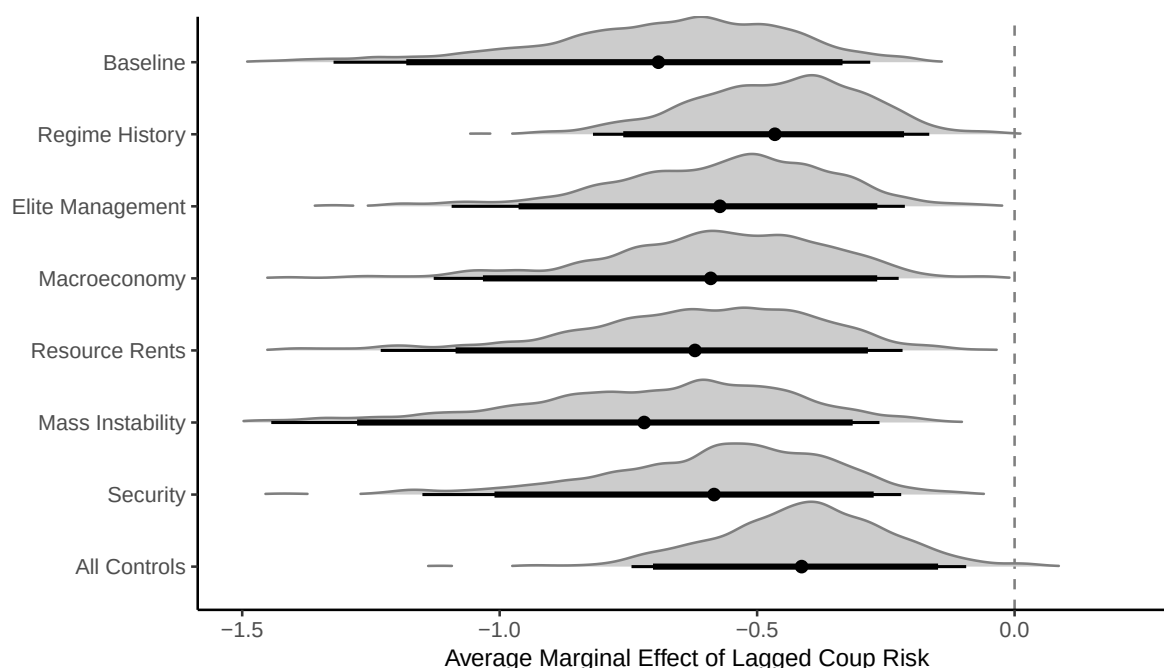
Notes: Black lines show the density of observed values. Grey lines show the density of imputed values for each of the five imputed datasets. Imputation uses predictive mean matching via mice.

Table B2. Observed vs. Imputed Values for Most-Missing Variables

Variable	Observed		Imputed		N_{miss}
	Mean	SD	Mean	SD	
Gas Rents	0.067	0.344	0.031	0.209	537
Coal Rents	0.361	3.649	0.689	2.791	486
Oil Rents	3.909	10.182	2.455	7.771	474
Leg. Frac.	0.159	0.241	0.271	0.291	408
Metal Rents	0.945	2.814	0.669	1.951	292
Mil. Exp.	217584.066	563837.076	192544.902	558805.562	203

Notes: Imputed values pooled across 5 imputed datasets. Imputation uses predictive mean matching via mice.

Figure B3. Average Marginal Effect of Coup Probability on Having Succession Rules in Complete-Case Sample



Notes: Distributions represent results from 1,000 clustered bootstrap samples. Thick lines represent 90% confidence intervals and thin lines 95% confidence intervals. Complete-case sample excludes observations with any missing predictor values.

Figure B3 replicates the main analysis using only complete cases, or observations with no missing values on any predictor variable. The complete-case sample contains 915 country-years (48% of the full 1,909-observation sample), with the reduction driven primarily by missing resource rent and legislative fractionalization data. Despite this reduction, the average marginal effect of coup probability on having succession rules remains negative and significant across all eight model specifications, with magnitudes comparable to the main results. Table B3 provides the full regression results.

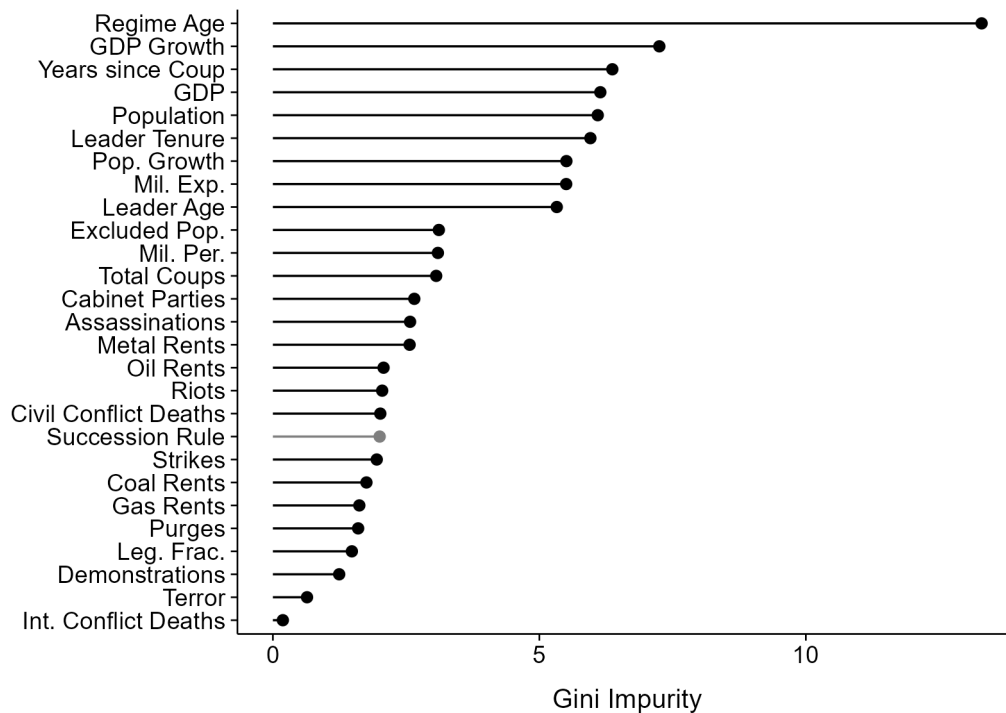
Table B3. Regression Results for Complete-Case Probit Models of Succession Rules

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Coup Risk	-2.754*** (1.155)	-2.424*** (0.945)	-2.592*** (1.068)	-2.588*** (1.154)	-2.630*** (1.176)	-2.956*** (1.331)	-2.669*** (1.106)	-2.962*** (1.312)
Regime Age		-0.010 (0.016)						-0.011 (0.021)
Total Coups		-0.080 (0.089)						-0.095 (0.157)
Years since Coup		0.058*** (0.021)						0.075*** (0.031)
Leader Age			0.026 (0.017)					-0.009 (0.023)
Cabinet Parties			0.146 (0.124)					0.174 (0.186)
Leg. Frac.			0.789 (0.734)					1.150 (1.178)
Excluded Pop.			-0.337 (0.550)					0.185 (0.764)
Logged GDP				0.405 (0.232)				0.326 (0.424)
GDP Growth				1.067 (1.105)				2.571 (1.333)
Logged Population				-0.495*** (0.281)				-0.425 (0.410)
Pop. Growth				-4.652 (19.002)				-2.655 (24.997)
Oil Rents					0.047 (0.323)			0.050 (0.634)
Gas Rents					69.655 (1398.372)			87.312 (2155.550)
Coal Rents					49.608 (516.494)			49.176 (489.818)
Metal Rents					0.126 (0.237)			0.142 (0.329)
Demonstrations						0.103 (0.124)		0.048 (0.190)
Purges						-0.165 (0.243)		0.058 (0.420)
Riots						0.003 (0.130)		-0.006 (0.208)
Strikes						0.387 (0.452)		0.086 (0.626)
Terror						-0.350 (0.402)		0.042 (0.515)
Logged Mil. Exp.							0.166*** (0.109)	0.090 (0.151)
Mil. Per.							-0.010 (0.006)	-0.010 (0.011)
Logged Civil Deaths							-0.002 (0.051)	-0.011 (0.091)
Logged Int. Deaths							0.014 (0.373)	0.085 (0.369)
Intercept	0.990*** (0.181)	0.622 (0.472)	-0.595 (0.920)	-1.759 (1.997)	0.813*** (0.221)	1.018*** (0.175)	-0.456 (1.184)	-2.386 (3.425)
N	875	875	875	875	875	875	875	875

Notes: *** 95% confidence interval based on 1,000 clustered bootstrapped samples excludes 0.
All independent variables lagged by one year.

C Reverse Causality

Figure C1. Importance of Variables in the Random Forest Model for Predicting Coups with Succession Rules as a Predictor



Notes: Variables are ranked according to Gini impurity. Variables with a higher reduction in Gini impurity are more important to the Random Forest model because they are used to split nodes more frequently.

I address the possibility of reverse causality using two strategies. First, I re-estimate the Random Forest model with succession rules included as an additional predictor. If succession rules are a strong predictor of coups, then the negative correlation in the main analysis could reflect rules reducing coup risk rather than coup risk discouraging rules. Figure C1 displays the variable importance rankings from this amended model. Succession rules rank in the bottom third of predictors, comparable to variables such as strikes and coal rents. The model's area under the precision-recall curve improves only marginally, from 0.254 to 0.268, and the predicted coup probabilities from the two models correlate at an R^2 of 0.931. There is little evidence that succession rules add meaningful predictive information about coups.

Second, I estimate bivariate probit models that jointly model coups as a function of succession rules and succession rules as a function of coup probability. Table C1 reports results from two specifications. Model 1 estimates the coup equation with succession rules and controls but without lagged coup probability. Model 2 adds lagged coup probability to the coup equation. In model 1, the coefficient on succession rules in the coup equation is negative and significant in seven of eight specifications, consistent with existing arguments that succession rules reduce coups. However, in model 2, which controls for baseline coup risk, the coefficient on succession rules reverses sign, becoming positive in all eight specifications and significant in five. This pattern is consistent with the argument that succession rules can themselves provoke coup attempts by empowering successors and angering disadvantaged elites. Across both specifica-

Table C1. Bivariate Probit Models

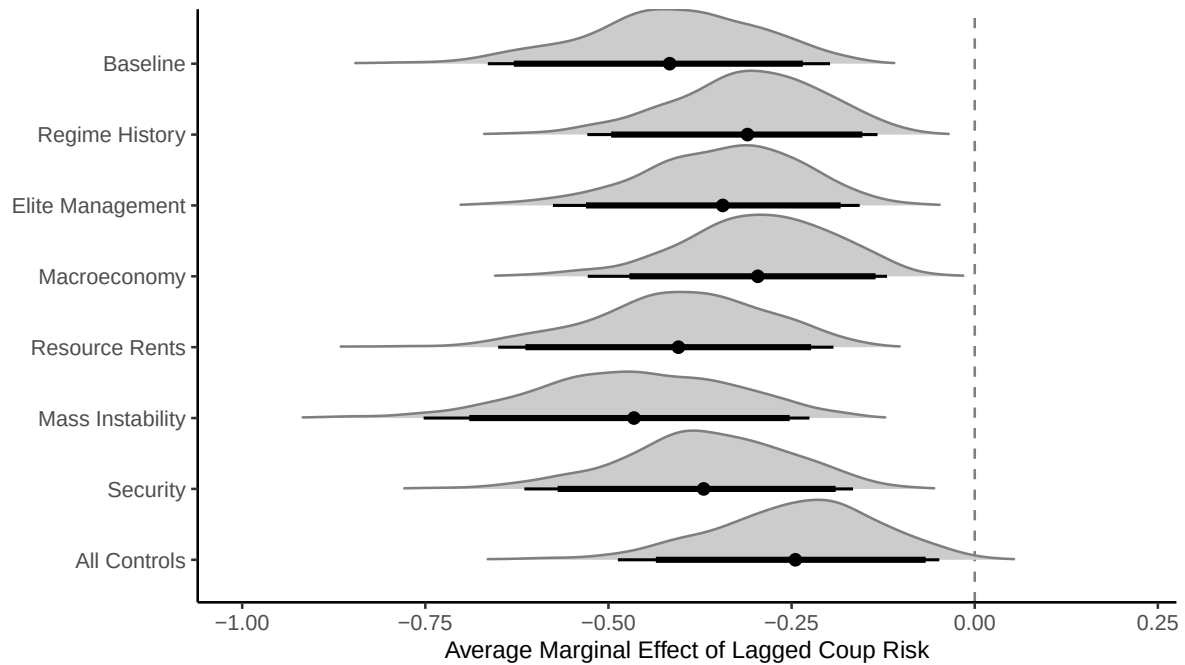
	Model 1			Model 2		
	$\beta(\text{Risk})$	$\alpha(\text{Rule})$	$\hat{\rho}$	$\beta(\text{Risk})$	$\alpha(\text{Rule})$	$\hat{\rho}$
(1) Baseline	-2.908*** (0.673)	-1.874*** (0.293)	0.727	-2.382*** (0.816)	1.461*** (0.442)	-0.879
(2) Regime History	-2.793*** (0.632)	-1.583*** (0.354)	0.627	-2.471*** (0.718)	1.211 (0.584)	-0.728
(3) Elite Management	-2.675*** (0.625)	-1.259*** (0.428)	0.511	-2.424*** (0.651)	0.581 (0.506)	-0.381
(4) Macroeconomy	-2.683*** (0.673)	-1.347*** (0.400)	0.533	-2.467*** (0.716)	1.015*** (0.427)	-0.664
(5) Resource Rents	-2.862*** (0.666)	-1.738*** (0.313)	0.698	-2.556*** (0.743)	1.165*** (0.505)	-0.737
(6) Mass Instability	-3.062*** (0.751)	-1.769*** (0.294)	0.702	-2.653*** (0.849)	1.303*** (0.504)	-0.797
(7) Security	-2.827*** (0.665)	-1.662*** (0.372)	0.629	-2.566*** (0.692)	1.105*** (0.422)	-0.744
(8) All Controls	-2.682*** (0.782)	-0.855 (0.440)	0.367	-2.536*** (0.802)	0.441 (0.458)	-0.239

Notes: Bivariate probit estimated jointly via maximum likelihood. $N = 1937$. Rule equation: Rule_{t-1} regressed on predicted coup probability at $t - 2$ and controls at $t - 2$. Model 1 coup equation: Coup_t regressed on Rule_{t-1} and controls at $t - 1$. Model 2 coup equation adds predicted coup probability at $t - 1$ to the coup equation. $\hat{\rho}$ is the estimated residual correlation. Point estimates and standard errors from cluster bootstrap with $B = 1000$. *** denotes 95% confidence interval excludes zero.

tions, the coefficient on coup probability in the rule equation remains negative and significant, consistent with the main results.

D Unobserved Heterogeneity

Figure D1. Average Marginal Effect of Coup Probability on Having Succession Rules with Correlated Random Effects



Notes: Distributions represent results from 1,000 clustered bootstrap samples. Thick lines represent 90% confidence intervals and thin lines 95% confidence intervals.

The probit models in the main analysis do not account for time-invariant unobserved heterogeneity across autocratic spells. If unobserved characteristics of specific autocratic spells—such as features of the founding regime or the political culture—simultaneously affect both coup risk and the presence of succession rules, the main estimates could be biased. To address this concern, I re-estimate the models using correlated random effects, which Crisman-Cox (2021) identifies as among the best-performing estimators for imbalanced binary panel data. Figure D1 displays the average marginal effect of coup probability on having succession rules from the correlated random effects models. The point estimates are attenuated relative to the main results, as expected when accounting for unit-level heterogeneity, but remain negative and significant across all eight specifications.

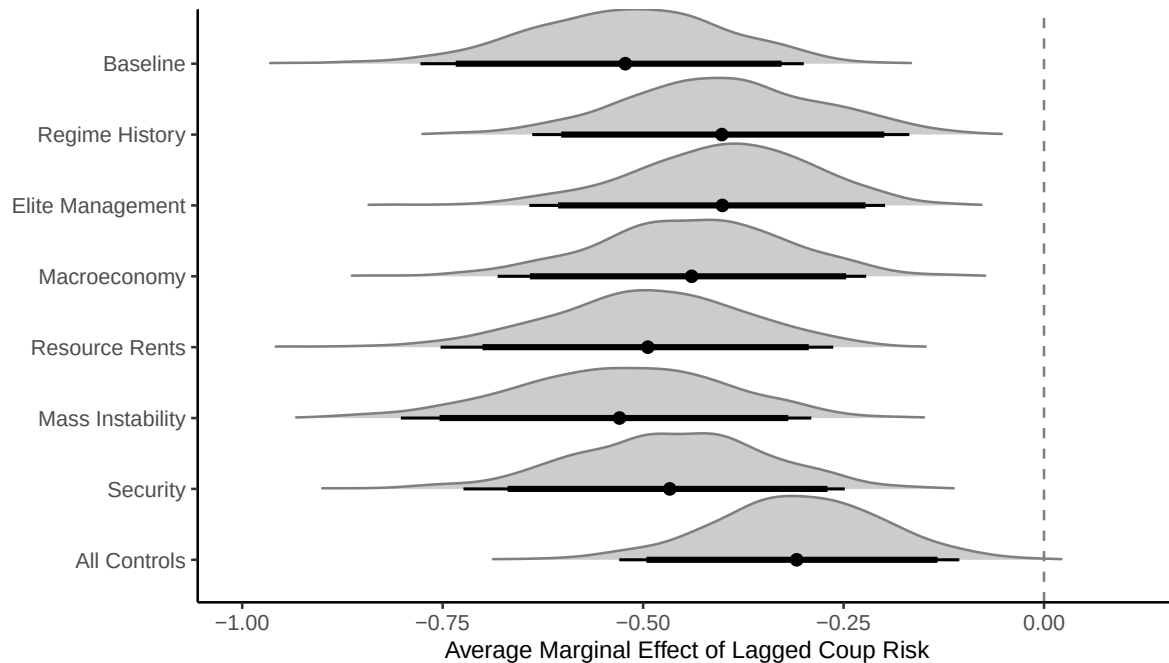
Table D1. Regression Results for CRE Probit Models of Succession Rules

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Coup Risk	-1.706*** (0.526)	-1.563*** (0.517)	-1.710*** (0.564)	-1.414*** (0.508)	-1.756*** (0.544)	-2.075*** (0.634)	-1.736*** (0.528)	-2.385*** (1.185)
Regime Age		0.026 (0.013)						0.038 (0.042)
Total Coups		-0.039 (0.155)						-0.024 (0.258)
Years since Coup		0.020*** (0.012)						0.027 (0.027)
Leader Age			0.028*** (0.011)					0.036*** (0.020)
Cabinet Parties			0.156 (0.126)					0.139 (0.202)
Leg. Frac.			0.281 (0.389)					0.519 (0.709)
Excluded Pop.			-1.034*** (0.525)					-1.304 (1.047)
Logged GDP				0.279 (0.214)				0.272 (0.560)
GDP Growth				0.684 (0.531)				0.753 (1.098)
Logged Population				0.493 (0.364)				-0.165 (1.089)
Pop. Growth				-10.328 (10.799)				-14.179 (15.888)
Oil Rents					0.005 (0.060)			-0.009 (0.130)
Gas Rents					1.233 (2.654)			2.242 (17.145)
Coal Rents					0.687 (6.285)			0.642 (8.225)
Metal Rents					0.055 (0.077)			0.083 (0.148)
Demonstrations						0.145 (0.106)		0.077 (0.191)
Purges						-0.166 (0.118)		-0.113 (0.200)
Riots						0.042 (0.079)		0.000 (0.177)
Strikes						0.279 (0.372)		0.350 (0.626)
Terror						0.115 (0.111)		0.245 (0.218)
Logged Mil. Exp.							0.058*** (0.025)	-0.040 (0.068)
Mil. Per.							0.002 (0.003)	-0.003 (0.010)
Logged Civil Deaths							-0.038 (0.040)	-0.152*** (0.064)
Logged Int. Deaths							0.051 (0.139)	0.178 (0.228)
Intercept	1.277*** (0.217)	0.949 (0.533)	0.203 (1.390)	-2.837 (1.941)	1.016*** (0.299)	1.349*** (0.271)	-0.600 (1.308)	-15.857 (25.725)
N	1862	1862	1862	1862	1862	1862	1862	1862

Notes: *** 95% confidence interval based on 1,000 clustered bootstrapped samples excludes 0.
Standard errors based on 1,000 clustered bootstrapped samples in parentheses.
All independent variables lagged by one year. Mundlak means are included but not shown.

E Coup Attempts

Figure E1. Average Marginal Effect of Coup Probability on Having Succession Rules Using Coup Attempt Probability



Notes: Distributions represent results from 1,000 clustered bootstrap samples. Thick lines represent 90% confidence intervals and thin lines 95% confidence intervals.

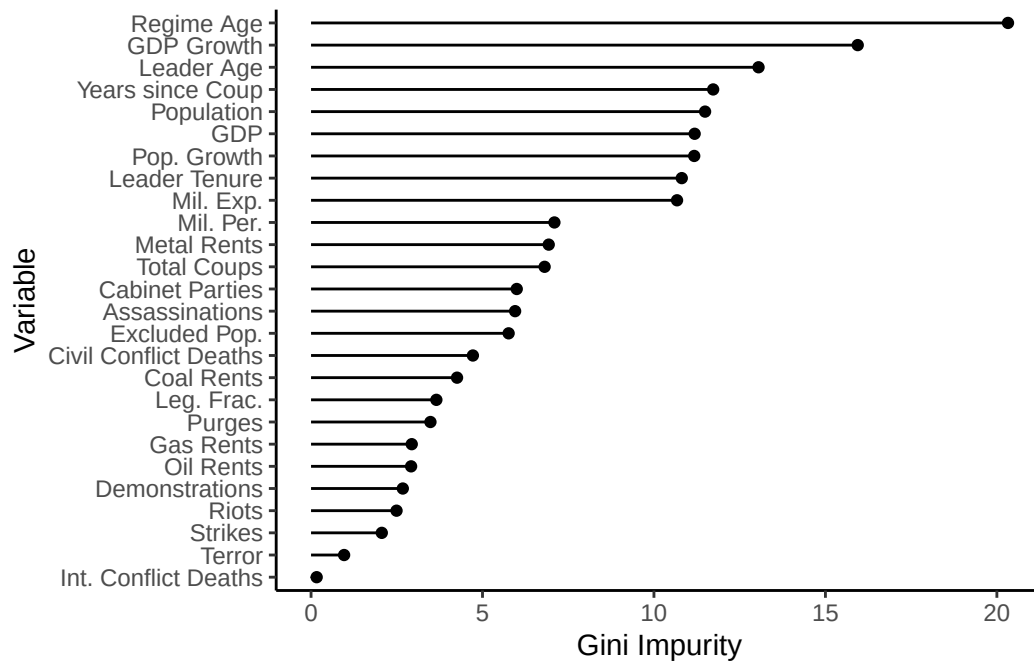
The main analysis predicts only successful coups on the grounds that dictators are most threatened when coups can credibly succeed. However, dictators may also wish to avoid coup attempts regardless of their outcome, as even failed attempts can destabilize a regime and signal vulnerability. To assess whether the results are sensitive to this distinction, I replicate the full analysis using the predicted probability of any coup attempt—successful or failed—as the independent variable. Figure E1 displays the average marginal effect. The results are substantively unchanged. The AME of coup attempt probability on having succession rules is negative and significant in all eight model specifications.

Table E1. Regression Results for Coup Attempt Probit Models of Succession Rules

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Coup Risk	-2.077*** (0.536)	-1.743*** (0.532)	-1.840*** (0.544)	-1.896*** (0.540)	-2.027*** (0.554)	-2.144*** (0.594)	-2.021*** (0.550)	-1.794*** (0.654)
Regime Age		-0.005 (0.011)						-0.003 (0.010)
Total Coups		-0.035 (0.062)						-0.092 (0.086)
Years since Coup		0.024*** (0.012)						0.015 (0.014)
Leader Age			0.023*** (0.010)					0.010 (0.015)
Cabinet Parties			0.270*** (0.131)					0.335*** (0.160)
Leg. Frac.			0.151 (0.455)					0.153 (0.636)
Excluded Pop.			-0.759 (0.422)					-0.804*** (0.407)
Logged GDP				0.429*** (0.154)				0.408*** (0.204)
GDP Growth				0.412 (0.710)				0.676 (0.924)
Logged Population				-0.429*** (0.164)				-0.364*** (0.175)
Pop. Growth				-2.623 (11.295)				-2.532 (13.136)
Oil Rents					0.014 (0.069)			-0.003 (0.040)
Gas Rents					1.827 (2.888)			1.285 (2.693)
Coal Rents					0.610 (3.424)			0.451 (3.285)
Metal Rents					0.087 (0.083)			0.080 (0.096)
Demonstrations						0.184*** (0.093)		0.131 (0.102)
Purges						-0.229 (0.137)		-0.179 (0.137)
Riots						-0.000 (0.086)		-0.012 (0.110)
Strikes						0.179 (0.311)		0.184 (0.357)
Terror						0.065 (0.076)		0.186*** (0.127)
Logged Mil. Exp.							0.102*** (0.041)	0.068 (0.058)
Mil. Per.							-0.006 (0.003)	-0.005 (0.004)
Logged Civil Deaths							-0.040 (0.035)	-0.070 (0.038)
Logged Int. Deaths							0.043 (0.166)	0.072 (0.169)
Intercept	1.064*** (0.163)	0.918*** (0.364)	-0.354 (0.534)	-2.112 (1.331)	0.937*** (0.182)	1.039*** (0.164)	0.251 (0.481)	-3.321*** (1.533)
N	1862	1862	1862	1862	1862	1862	1862	1862

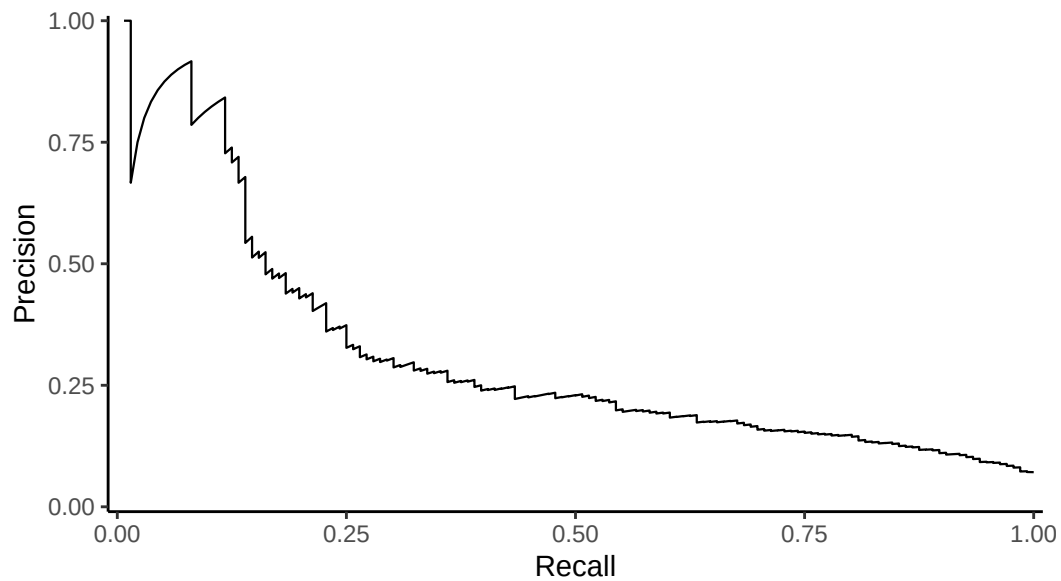
Notes: *** 95% confidence interval based on 1,000 clustered bootstrapped samples excludes 0.
Standard errors based on 1,000 clustered bootstrapped samples in parentheses.
All independent variables lagged by one year.

Figure E2. Importance of Variables in the Random Forest Model for Predicting Coup Attempts



Notes: Variables are ranked according to Gini impurity. Variables with a higher reduction in Gini impurity are more important to the Random Forest model because they are used to split nodes more frequently.

Figure E3. Precision Recall Curve for the Random Forest Model for Predicting Coup Attempts



Notes: $\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$. $\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$. The Random Forest trained on coup attempts has an area under the curve of 0.303, 4.3 times better than a random classifier (baseline AUC = 0.071).

F Succession Rules in Zimbabwe

Directly observing the internal dynamics of autocratic succession is challenging. Autocracies are generally more secretive than their democratic counterparts, particularly in matters of high-level politics between the dictator and elites (Barros 2016). Succession is an especially sensitive issue, to the extent that dictators often obscure their succession plans even from elites. Although constitutions are public, the process for drafting them often is not. The evidentiary record on drafting constitutions is particularly lacking in sub-Saharan Africa, home to many of the modern world's autocratic states.

Zimbabwe is an example where changes in succession rules coincided with well-documented shifts in dictator Robert Mugabe's control over the regime. Zimbabwe's first succession rules reflected path dependence from British rule. When Zimbabwe changed its succession rule to reflect the underlying power dynamics, the dictator was at the peak of his power and the succession rule was formalized with a specific successor. The Zimbabwean case shows, consistent with my argument, that a more powerful dictator formalized succession rules whereas previous arguments anticipate that a strengthened dictator would lack the incentives to formalize such a rule.

Initially, Zimbabwe lacked a formal succession rule but introduced one later. Zimbabwe and its predecessor states, Rhodesia and Zimbabwe Rhodesia, had parliamentary systems with ceremonial presidents. In Rhodesia, the cabinet appointed the president, or the "Officer Administering the Government." The presidency was intended to be a temporary office until the United Kingdom recognized Rhodesia's independence and the queen's representative appointed a prime minister, a change that never occurred. Parliament elected the ceremonial president in the short-lived Zimbabwe Rhodesia. Neither Rhodesia nor Zimbabwe Rhodesia required formal succession systems for the prime minister as chief executive because the selection process could simply be repeated if a vacancy occurred.

After Zimbabwe gained independence, the country remained parliamentary. Robert Mugabe ruled as prime minister, and the legislature elected a ceremonial president. As before, prime ministerial vacancies were addressed by repeating the selection process. A constitutional amendment in 1987 restructured the government, centralizing power in the presidency and eliminating the position of prime minister. Mugabe, who drove the reforms, assumed the presidency. Along with creating an executive presidency, the amendment introduced a vice-president, appointed and dismissible by the president. After a vacancy, the vice-president would become acting president until new elections were held within 90 days to formally begin a new term for the new president, presumably chosen informally by the ruling party. A 1990 constitutional amendment added a second vice-president, placed beneath the first vice-president in the line of succession.

Consistent with my argument, the constitutional reforms came at the height of Mugabe's power and safety, not at a moment of weakness as anticipated by previous work. In 1987, Mugabe successfully ended a campaign to violently suppress the opposition Zimbabwe African People's Union and merge its remaining members into the new ruling party, the Zimbabwe African National Union-Patriotic Front (ZANU-PF). These reforms further empowered Mugabe, granting him nearly all formal executive powers, including almost complete control of the cabinet (Compagnon 2011). Mugabe's creation of the vice-presidency coincided with the peak of his power, rather than a period of weakness. As one biographer describes Mugabe after the 1987 amendment, Mugabe's "control of appointments to all senior posts in the civil service, the defence forces, the police, and parastatal organisations gave him a virtual stranglehold on

government machinery and unlimited opportunities to exercise patronage”(Meredith 2002, 79).

Zimbabwe’s introduction of a new constitution in 2013 also shows elite concerns over succession rules, particularly regarding their effects on a successor’s power and likely succession outcomes. In 2008, Mugabe reached an agreement with the opposition Movement for Democratic Change – Tsvangirai, which had won a plurality in the lower house, to introduce a new constitution. Succession rules were a key issue within the ZANU-PF. Mugabe was nearly 90 with no plans to retire. Barring a forced removal, Zimbabwe’s next president would likely emerge from Mugabe’s increasingly imminent death. The first draft of the new constitution included revisions to the presidential succession system. Under the proposed draft, both vice-presidents would be elected alongside the president on a joint ticket. If a vacancy occurred, the highest-ranking vice-president would fully assume the presidency for the entire remaining term.

High-ranking members of the ZANU-PF rejected the draft partially due to the succession provision. One concern was that giving Mugabe the power to name his running mates would be equivalent to letting him designate his successor. Mugabe, for his part, also wished to avoid making a definitive statement on a possible successor, which the running mate provision might force him to do. Another concern was that the new provisions would too strongly solidify Joice Mujuru, the incumbent first vice-president, as the most likely successor. Mujuru was a hero of Zimbabwe’s war for independence and kept a strong constituency through the ZANU-PF Women’s League. Although she was weakened politically by her husband’s death in a 2011 house fire, she remained likely to retain her position as vice-president. (Joice Mujuru’s husband, Solomon, was also a revolutionary hero and one of Zimbabwe’s top generals and richest men. Together, the Mujurus could form the most formidable challenge to Mugabe’s rule. Regional media implicated Mugabe in the fire that killed Solomon, and the fire has been described as “mysterious” with a “very meticulous nature... [that] suggested a professional disposal after a professional assassination” [Chan 2019, 230–31].) High-ranking officials in the ZANU-PF who opposed Mujuru’s succession feared that if she were named the running mate, she would be impossible to dislodge as the successor and might not wait for Mugabe’s death to seize power (Matyszak 2016).

As a compromise, the new constitution, approved by referendum in 2013, delayed the new succession system’s implementation for at least 10 years, almost certainly after Mugabe’s death or retirement. Schedule Six of the constitution established an interim system in which the president retained the authority to appoint and freely remove the vice-presidents. Upon a vacancy, the first vice-president would become acting president. The ZANU-PF, as the previous president’s party, would choose the next president. The interim succession system avoided giving any candidate a definitive advantage in succession. Instead, the party would negotiate the next leader only after Mugabe’s death.

Ultimately, the succession issue prevented Mugabe from holding power until his death. In 2014, Mugabe accused Mujuru of plotting a coup and removed her as first vice-president. Mugabe replaced Mujuru with her main rival in succession, Emmerson Mnangagwa. Grace Mugabe, the first lady, emerged as a contender for succession shortly thereafter. In 2017, Mugabe dismissed Mnangagwa as first vice-president, effectively establishing Grace as Mugabe’s hand-chosen successor. The military, however, favored Mnangagwa as the next president and forced Mugabe to resign two weeks later. The ZANU-PF invoked the Schedule Six succession rule and selected Mnangagwa as Mugabe’s successor (Chan 2019; Nyarota 2018; Tendi 2025). Although Mugabe eventually lost control of succession, the development of Zimbabwe’s succession rules illustrates key elements of my argument. Mugabe introduced a succession rule after he consolidated power, not when he needed to secure his rule. When Zimbabwe rewrote

the constitution, Mugabe and key officials in the ZANU-PF rejected the initial draft because it made the successor too strong and threatened Mugabe's hold on power.

G References

- Barros, Robert. 2016. "On the Outside Looking In: Secrecy and the Study of Authoritarian Regimes." *Social Science Quarterly* 97 (4): 953–73.
- Chan, Stephen. 2019. *Mugabe: A Life of Power and Violence*. 2nd ed. London: I. B. Tauris.
- Compagnon, Daniel. 2011. *A Predictable Tragedy: Robert Mugabe and the Collapse of Zimbabwe*. Philadelphia: University of Pennsylvania Press.
- Crisman-Cox, Casey. 2021. "Estimating Substantive Effects in Binary Outcome Panel Models: A Comparison." *Journal of Politics* 83 (2): 532–46.
- Matyszak, Derek. 2016. *Succession Politics and the ZANU PF Body Politic*. Harare, Zimbabwe: Southern African Political Economy Series.
- Meredith, Martin. 2002. *Mugabe: Power, Plunder, and the Struggle for Zimbabwe*. New York: PublicAffairs.
- Nyarota, Geoffrey. 2018. *The Graceless Fall of Robert Mugabe: The End of a Dictator's Reign*. Cape Town: Penguin Random House South Africa.
- Tendi, Blessing-Miles. 2025. *The Overthrow of Robert Mugabe: Gender, Coups, and Diplomats*. Oxford: Oxford University Press.